ISE 323: Human-Computer Interaction

Designing for Diversity
SBIAHU

Shubham Jain

Stony Brook University

Why design for diversity?

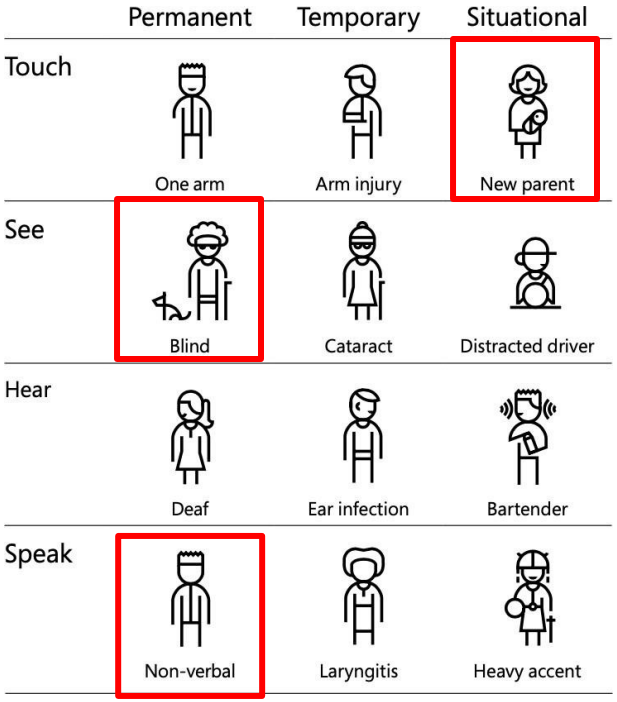
- It is important to design for the user, to follow a human-centered process
- However,
  - they have different abilities and weaknesses
  - they come from different backgrounds and cultures
  - they have different interests, viewpoints, and experiences
  - they are of different ages and sizes
  - …
- All these things have an impact on the way in which a person use a software application and, indeed, on whether they can use it at all

Are We Designing For People Like Us?

- If we use our own abilities as a starting point…
- … we make things that are easy for some people to use, but difficult for everyone else
- … we end up with systems designed for people like us
  - specific gender, age, language ability, tech literacy, physical ability, specific access to money, time, etc.

How do we define normal?

- The interactions we design with technology depend heavily on what we can understand/remember, see, hear, say, <u>and</u> touch
- Assuming all those senses and abilities are fully enabled all the time means ignoring several people
  - it also reflects how people really are, as “life happens”
- We want our designs to reflect that diversity



An Example of Universal Design

Take for example a news video online that has captions.

- **Situational**: You are on a loud train and have forgotten your headphones. You can still watch the news video and follow it by reading the video captions
- **Temporary**: You have just landed from a plane journey, your ears temporarily feel strange, and you can't hear properly. You can still watch the news video and follow it by reading the video captions
- **Permanent**: You may be hard of hearing or deaf. You can still watch the news video and follow it by reading the video captions

Principles and Methodologies

Inclusive and Universal Design

Inclusive Design

- A design methodology that enables and draws on the full range of human diversity
  - i.e., including and learning from people with a range of perspectives
- Designing a diversity of ways to participate so that everyone has a sense of belonging
- It not a "one size fits all" approach, but a "*one size fits one*"
  - it is more designing a system, a portion of it, or an application for a specific use case and extending this to others
- **Beware**: there is no "standard" and shared definitions, principles, and practices
  - here, we rely on a recent definition and practices by Microsoft Design (https://www.microsoft.com/design/inclusive/)

Microsoft's three fundamental principles of Inclusive Design

1. Recognize exclusion
2. Learn from Diversity
3. Solve for one, extend to many

Microsoft's three fundamental principles of Inclusive Design

1. **Recognize exclusion**
   - it means examining what you are building, and recognizing who would be excluded from using it
   - sometimes exclusion happens when we do not pay attention to our biases, and it could be temporary or situational
   - **Example:** if your app won't work well on older phones (e.g., for users who can only afford those), it will exclude them.

Microsoft's three fundamental principles of Inclusive Design

2. **Learn from diversity**
   - put people at the center of the design process from day zero
   - we can try to imagine how a person with a given set of abilities would use a system
   - we cannot imagine her various contexts, being them situational, emotional, or what gives her joy or frustration

3. **Solve for one, extend to many**
   - When you design a feature with one group in mind, you can expand the scope to help others who can benefit.
   - **Example:** If you offer users an option to listen to content rather than read it, you'll help users who have limited sight as well as those who may just want to rest their eyes.

The Beauty of Constraints

- Designing for people with permanent disabilities can seem like a significant constraint…
- … but the resulting designs can benefit a much larger number of people
- Examples?

Hard of hearing

The Beauty of Constraints

- Designing for people with permanent disabilities can seem like a significant constraint…
- … but the resulting designs can benefit a much larger number of people
- Examples
  - closed captioning was created for the hard of hearing community, but they are useful for reading in a crowded airport or to teach children how to read
  - remote controls, automatic door openers, audiobooks, …

The Beauty of Constraints

- **Example** – Ramps and Automatic Doors:
  - **Initial Purpose:** Designed for wheelchair users and those with mobility impairments.
  - **Wider Benefit:** Useful for parents with strollers, travelers with luggage, and delivery personnel.

Example: Creating a Video Game

- You are creating a video game for a **game console**
- A **competitive** game, with characters who needs to jump, run, … maybe even drive, at a certain stage.
- Who are you excluding?
  - Be sure to include various contexts/situations, not just "edge cases".
  - Which of them can you observe, and how?
- Which 1-2 solution(s) can we come to, then?

Example: Creating a Video Game

- Jumping, running, driving, … will require fine motor skills to compete.
- Possible factors to consider:
  - what if you have limited mobility?
  - or if you never played a video game before?
  - …

Example: Creating a Video Game

- **A possible solution:** a co-pilot mode
  - allows two game controllers to work together, so that two people can control the same character, or car, or…
  - in this way, an advanced or skilled player can play alongside someone who might need more assistance
- This opens gaming to various kinds of people
  - people with disabilities or temporary injuries
  - novice gamers
  - kids
  - people who just want to play together without competing

Copilot Mode – Xbox One

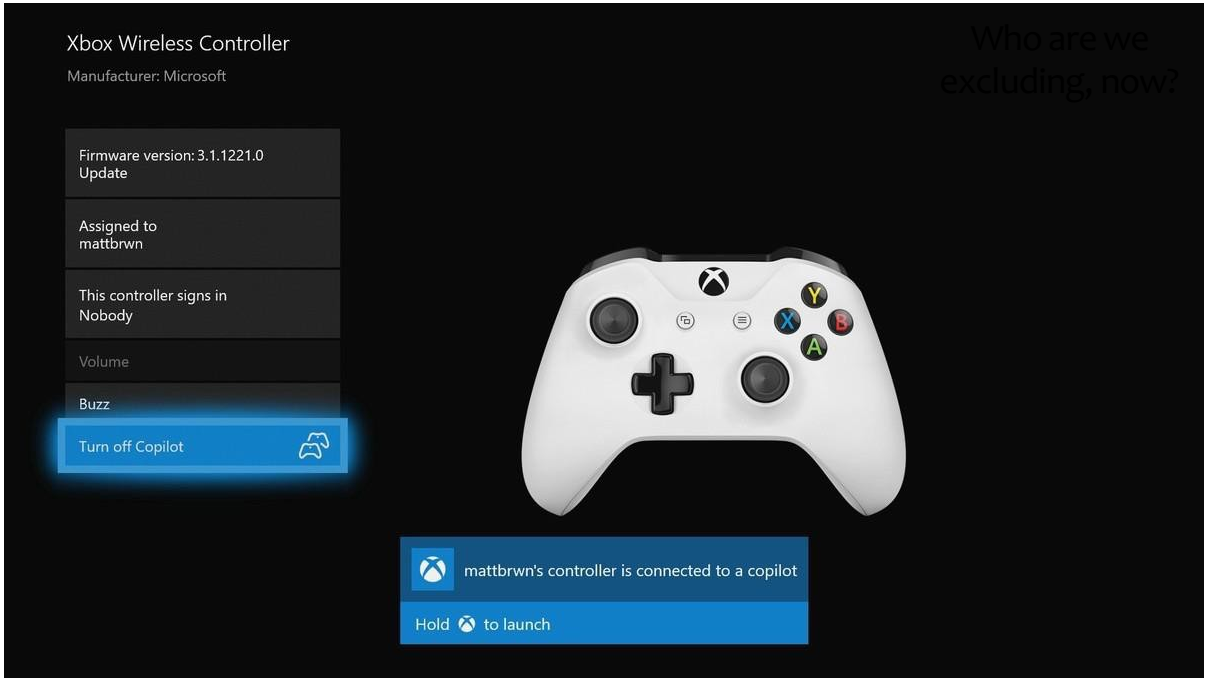
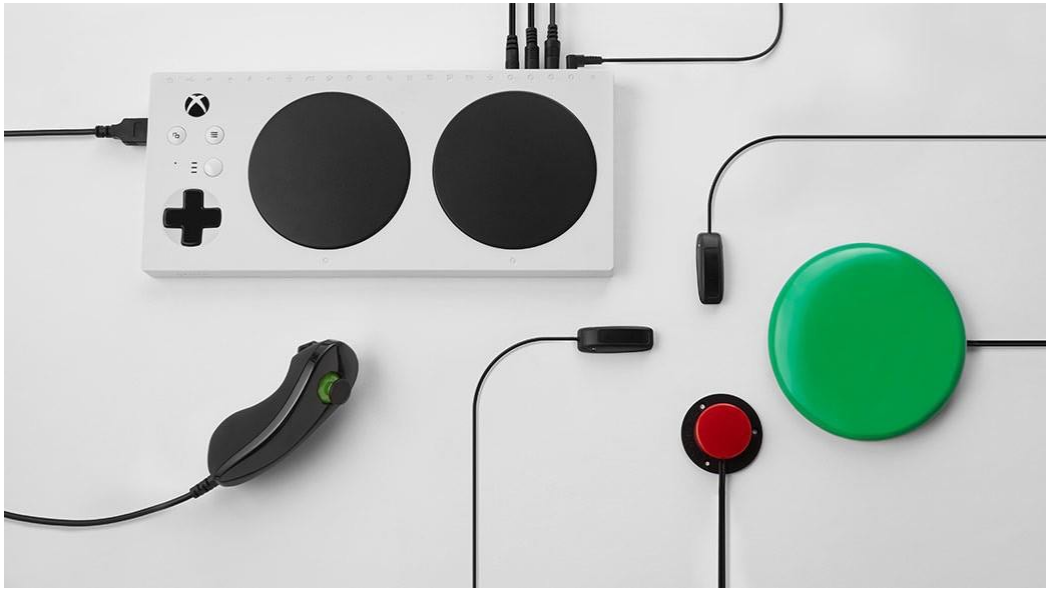


image from https://www.windowscentral.com/xbox-one-copilot

Xbox Adaptive Controller



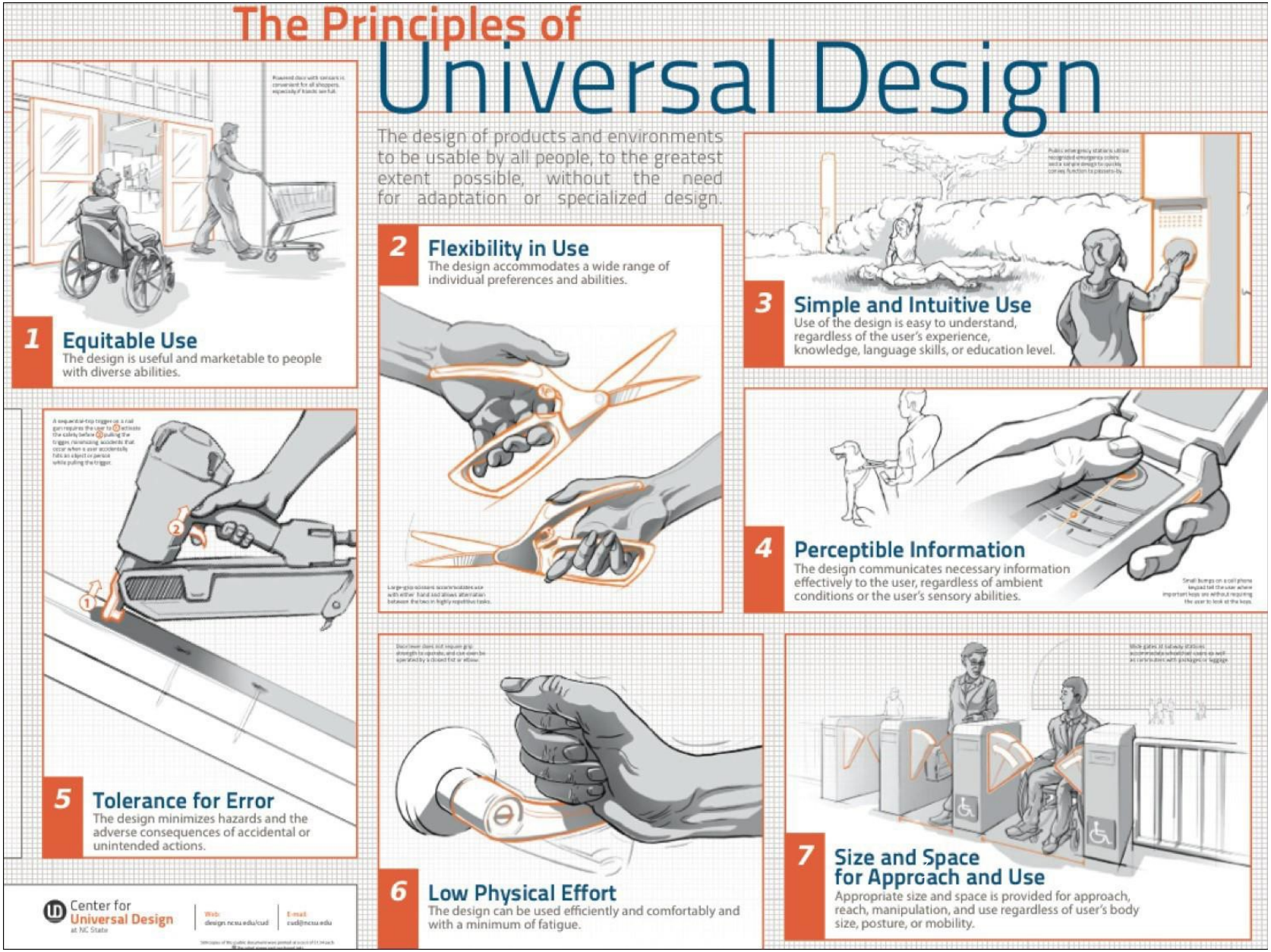
source: https://xbox.com/adaptive-controller

Are We Speaking About Accessibility?

- Not only
  - accessibility is an attribute, inclusive design is a method
- Accessibility focuses primarily on people with disabilities
  - ensuring that there are no barriers to serving them
  - via testable accommodations able to solve a technical, design, physical, or cognitive barrier to engaging with a system or product
- Inclusive design will make your systems and products more accessible, but it is not a process for meeting all accessibility standards
- Accessibility and inclusive design work together to make experiences that are not only compliant with some standards, but usable and open to all

Universal Design

- Designing interactive systems that are usable by anyone, with any range of abilities, using any technology platform
- A "one size fits all" approach, less prone to consider very specific cases
  - it is strongest at describing the qualities and nature of a final design
  - it might not involve the participation of some excluded communities
- Born for the physical world, then adopted in the digital one



https://www.interaction-design.org/literature/article/learn-to-create-accessible-websites-with-the-principles-of-universal-design

1. Equitable Use

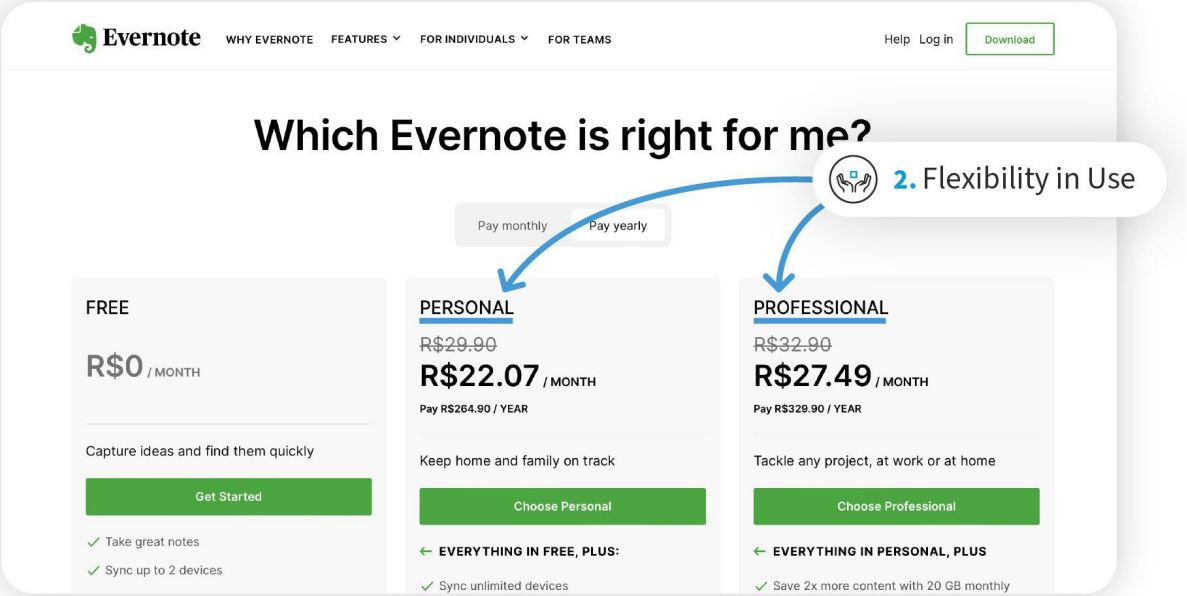
- Principle: The design is useful and marketable to people with diverse abilities.

- Example: Automatic doors at the entrance of a building. They can be used by everyone, including those who use wheelchairs, those carrying items, or parents pushing strollers.

2. Flexibility in Use

- Principle: The design accommodates a wide range of individual preferences and abilities.

Evernote's homepage offers flexibility in use via options to select. Users can approach Evernote in different contexts, for personal projects or for professional, team-based ones.



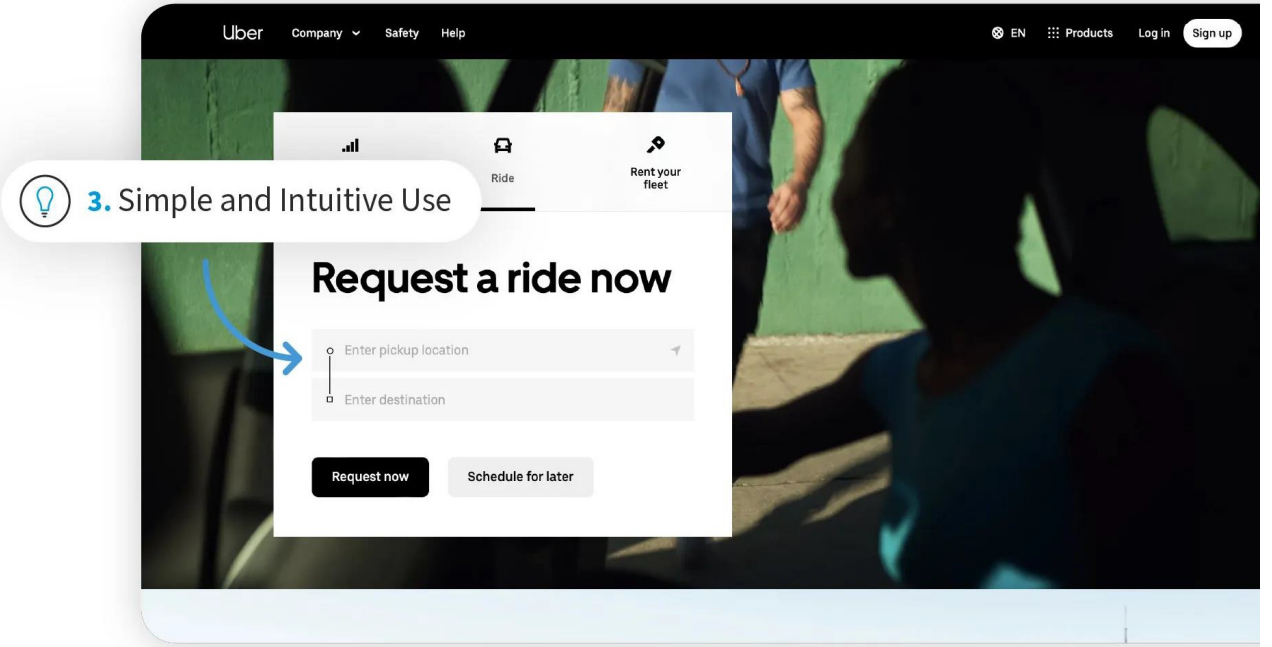
3. Simple and Intuitive Use

- Principle: Use of the design is easy to understand, regardless of the user's experience, knowledge, language skills, or current concentration level.

- Example: A universal TV remote with simple, clearly labeled buttons and an intuitive layout. Even those unfamiliar with the specific remote can easily operate it.

3. Simple and Intuitive Use

Uber’s ultra-easy Ride section offers users the quick convenience of pickup location and destination.



4. Perceptible Information

- Principle: The design communicates necessary information effectively to the user, regardless of ambient conditions or the user's sensory abilities.

- Example: An elevator that provides auditory announcements of each floor in addition to visual indicators. This ensures that both visually and hearing-impaired individuals receive the necessary information.

5. Tolerance for Error

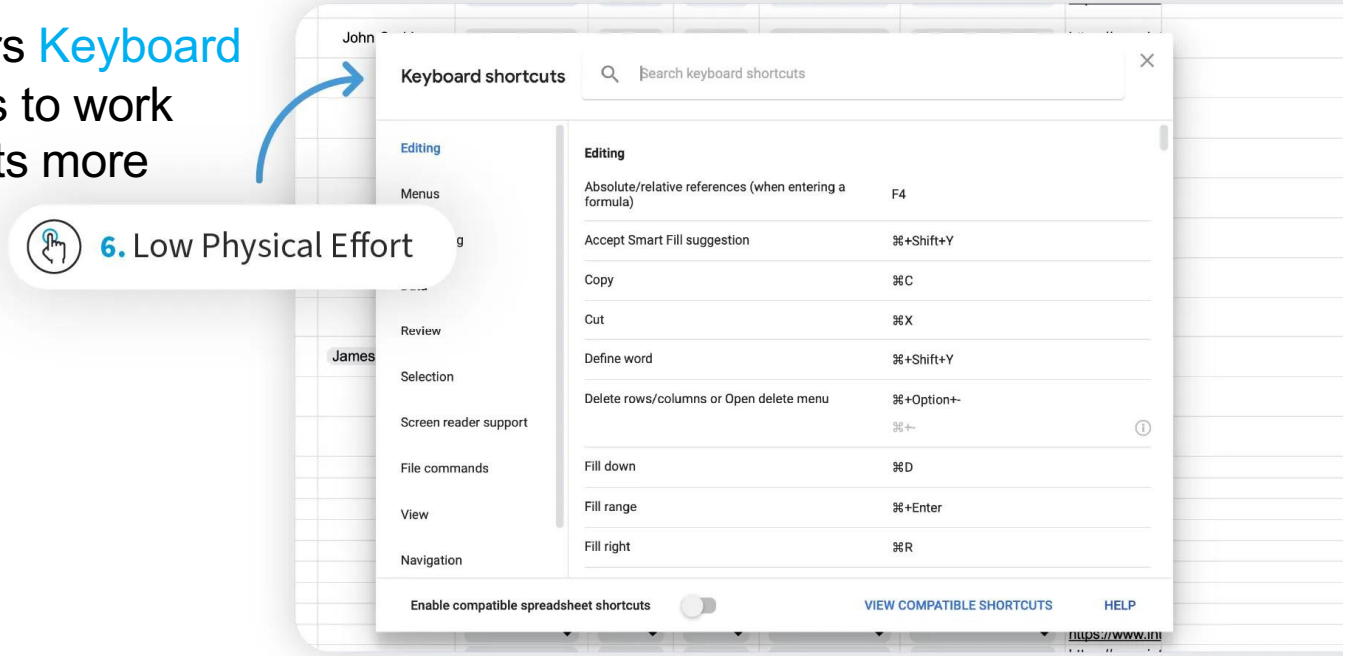
- Principle: The design minimizes hazards and the adverse consequences of accidental or unintended actions.

- Example: Software that automatically saves work periodically and allows users to undo recent actions. This helps prevent loss of data and reduces the impact of user errors.

6. Low Physical Effort

- Principle: The design can be used efficiently and comfortably and with a minimum of fatigue.

Google Docs offers Keyboard shortcuts for users to work with Google Sheets more efficiently.

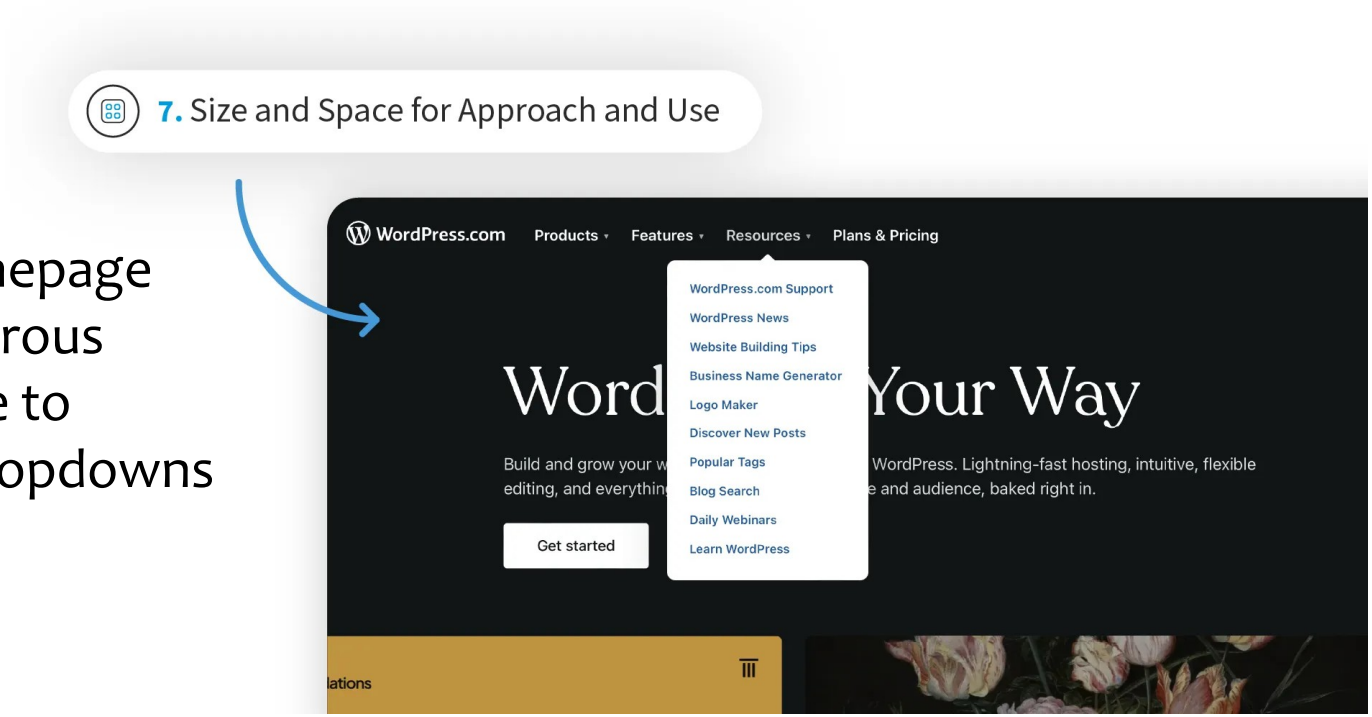


7. Size and Space for Approach and Use

- Principle: Appropriate size and space is provided for approach, reach, manipulation, and use regardless of user's body size, posture, or mobility.

- Example: A kitchen with adjustable countertops and cabinets that can be reached easily by both standing users and those in wheelchairs. This ensures that the kitchen is usable by individuals of various heights and mobility levels.

7. Size and Space for Approach and Use



WordPress's homepage offers users generous amounts of space to access desired dropdowns (at the top of the webpage).

Human Performance Modeling

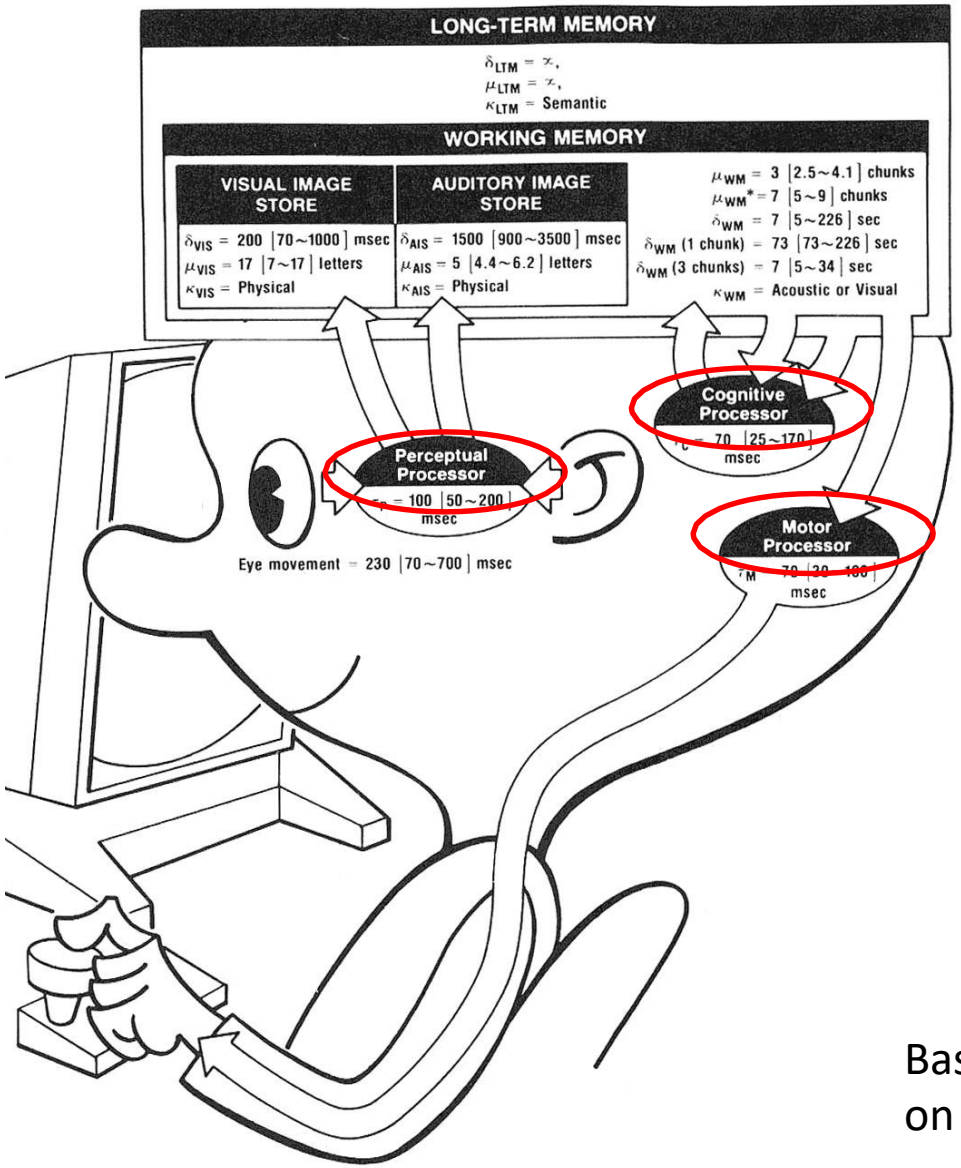
Motivation: The designer of the interface between human and machine needs to be able to analyze the tasks that the human will be asked to do and needs to be able to predict the human's performance.

A Simple HCI Task

- Pressing the corresponding key

B

Human Information Processor

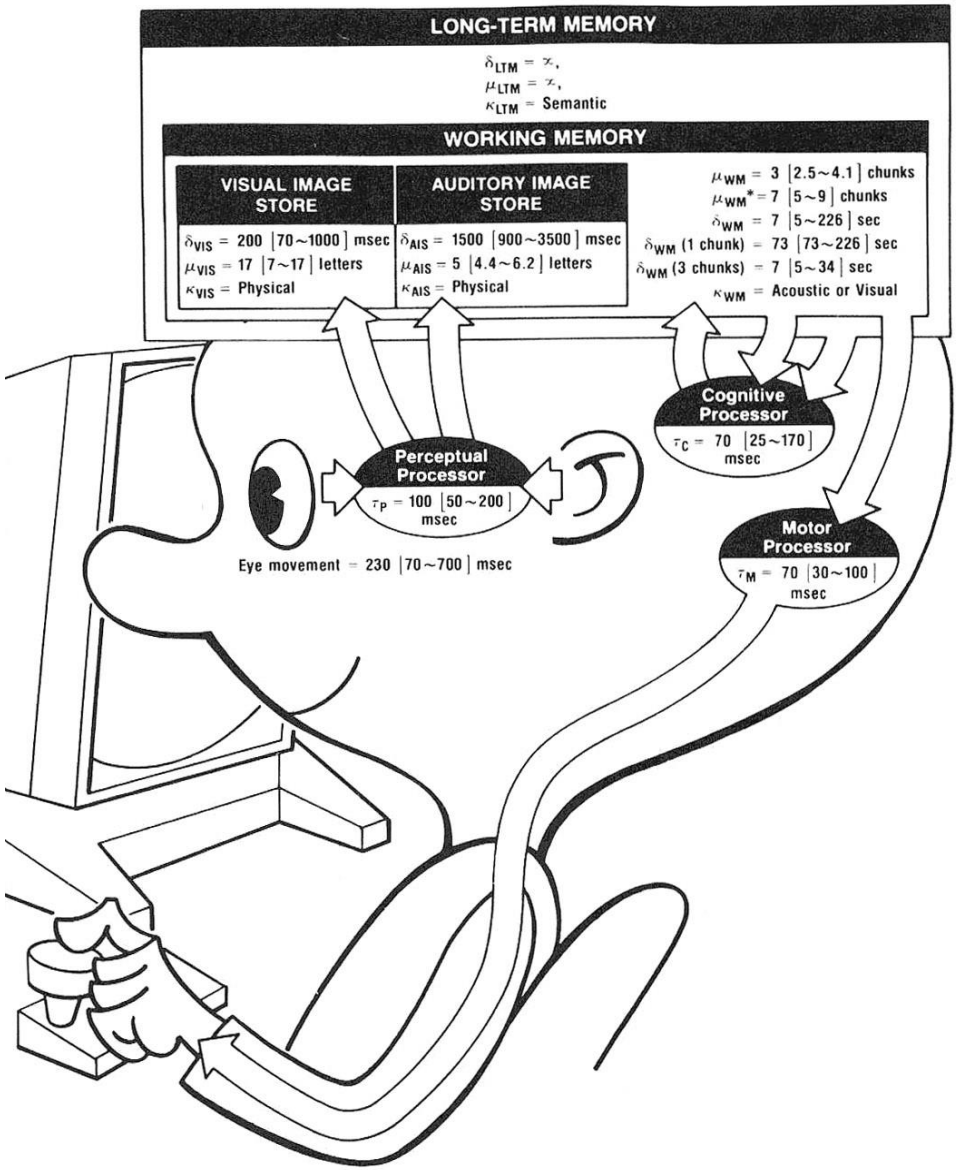


3 subsystems

- Perceptual
- Motor
- Cognitive

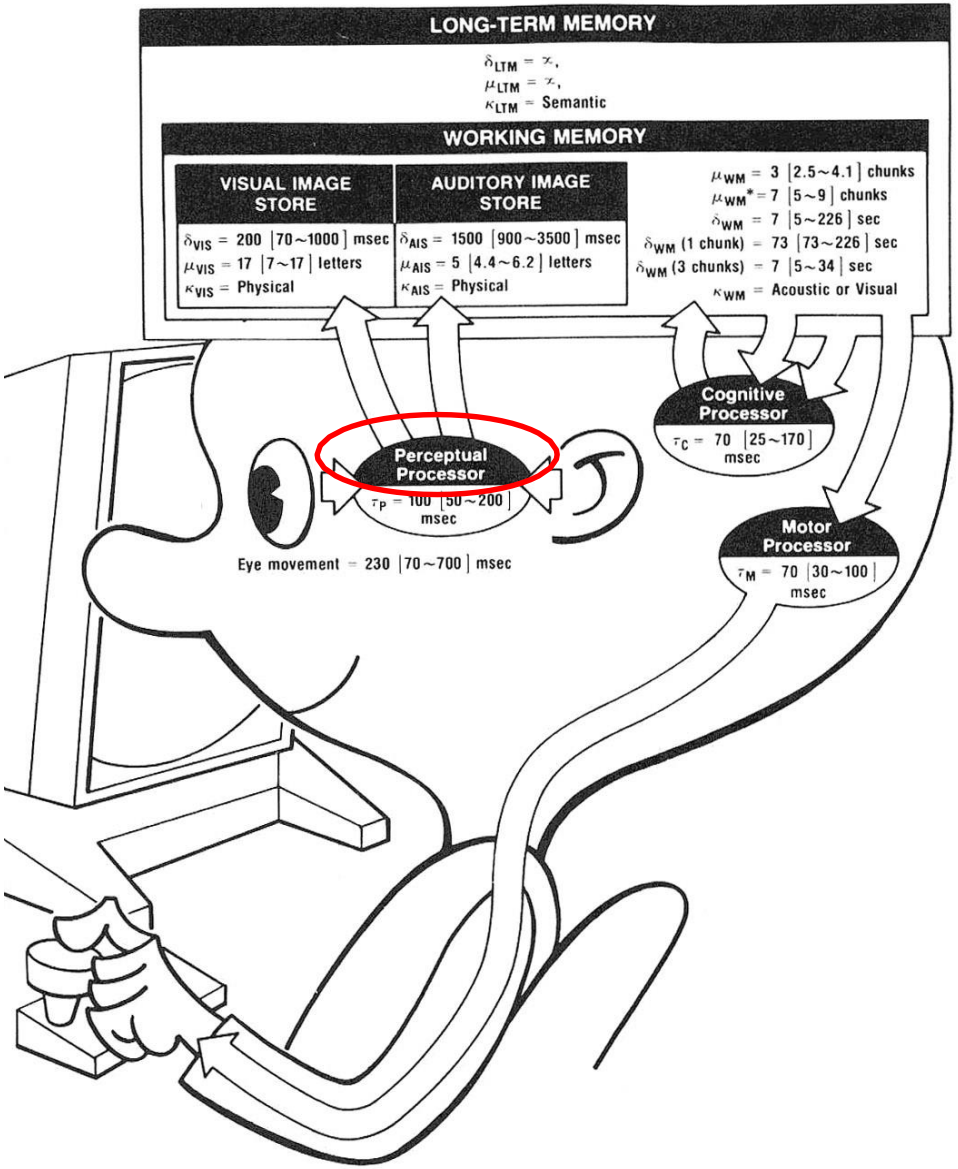
Based on Card, Moran and Newell's paper on the Human Processor Model

Human Information Processor



- Each subsystem has its own memories and processors.
- Memory
  μ, storage capacity in items
  δ, decay time of an item
- Processor
  τ, cycle time
- Three subsystems can work in parallel.

The Perceptual System



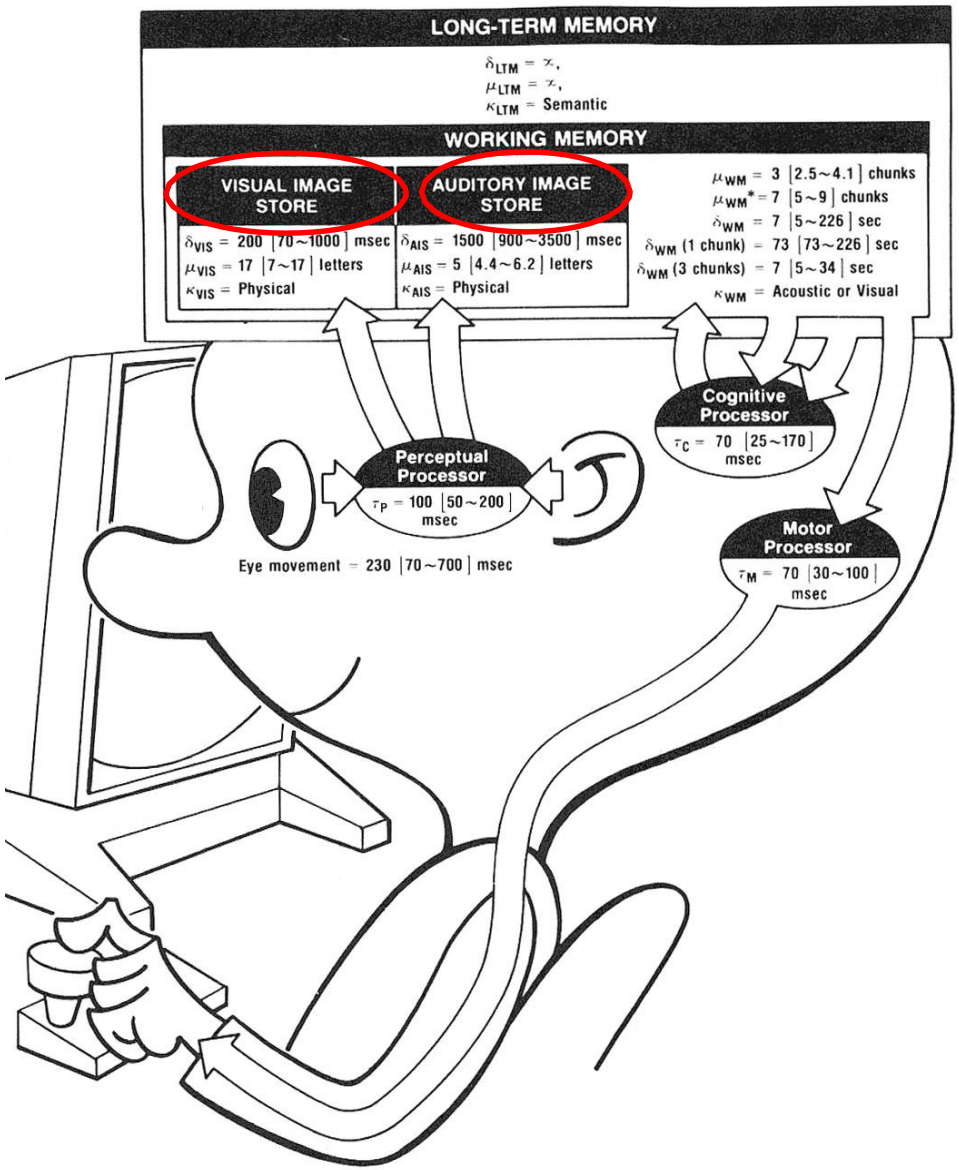
Function:

Carries the sensations of the physical world detected by the body's sensory systems into internal representations of the mind by means of integrated sensory systems.

- An Example: the visual system

  Eye-movement: 230 [70-700] ms

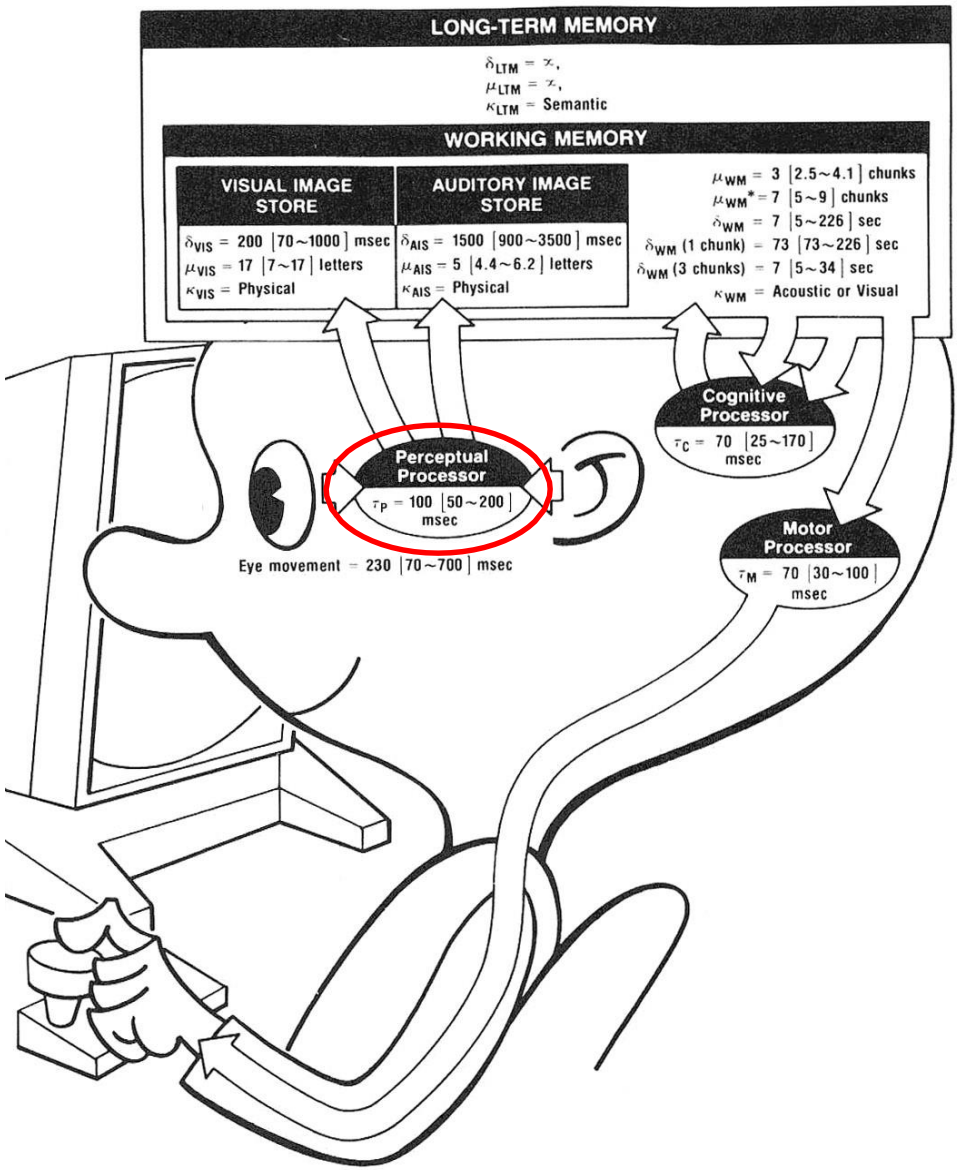
Perceptual Memories



δ_{VIS}=200[90~1000]ms
δ_{AIS}=1500[900~3500]ms

μ_{VIS}=17 [7~17] letters
μ_{AIS}=5 [4.4~6.2] letters

Perceptual Processor



τ_p=100[50~200] ms

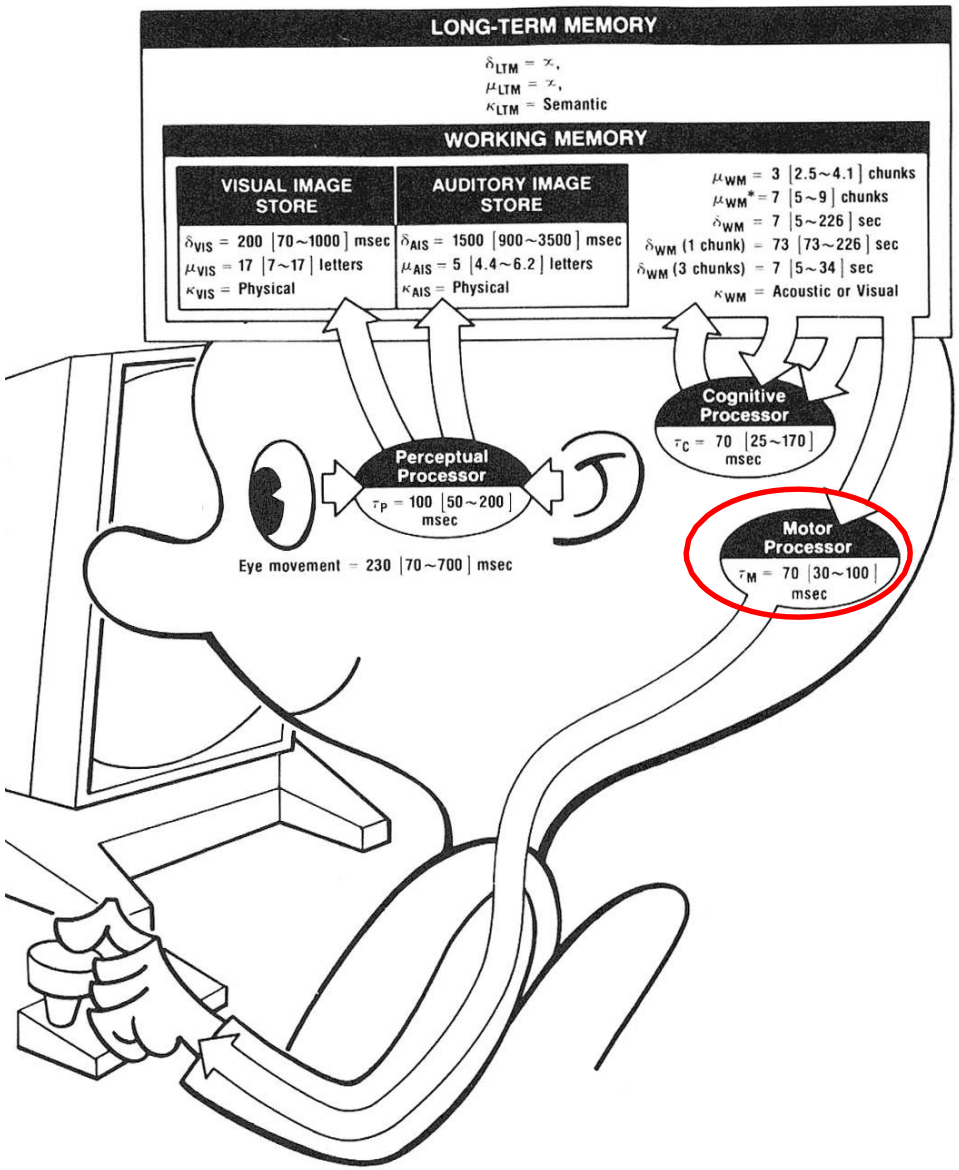
Perception

Moving Picture Rate

Example 1. Compute the frame rate at which an animated Image on a video display must be refreshed to give the illusion of movement.

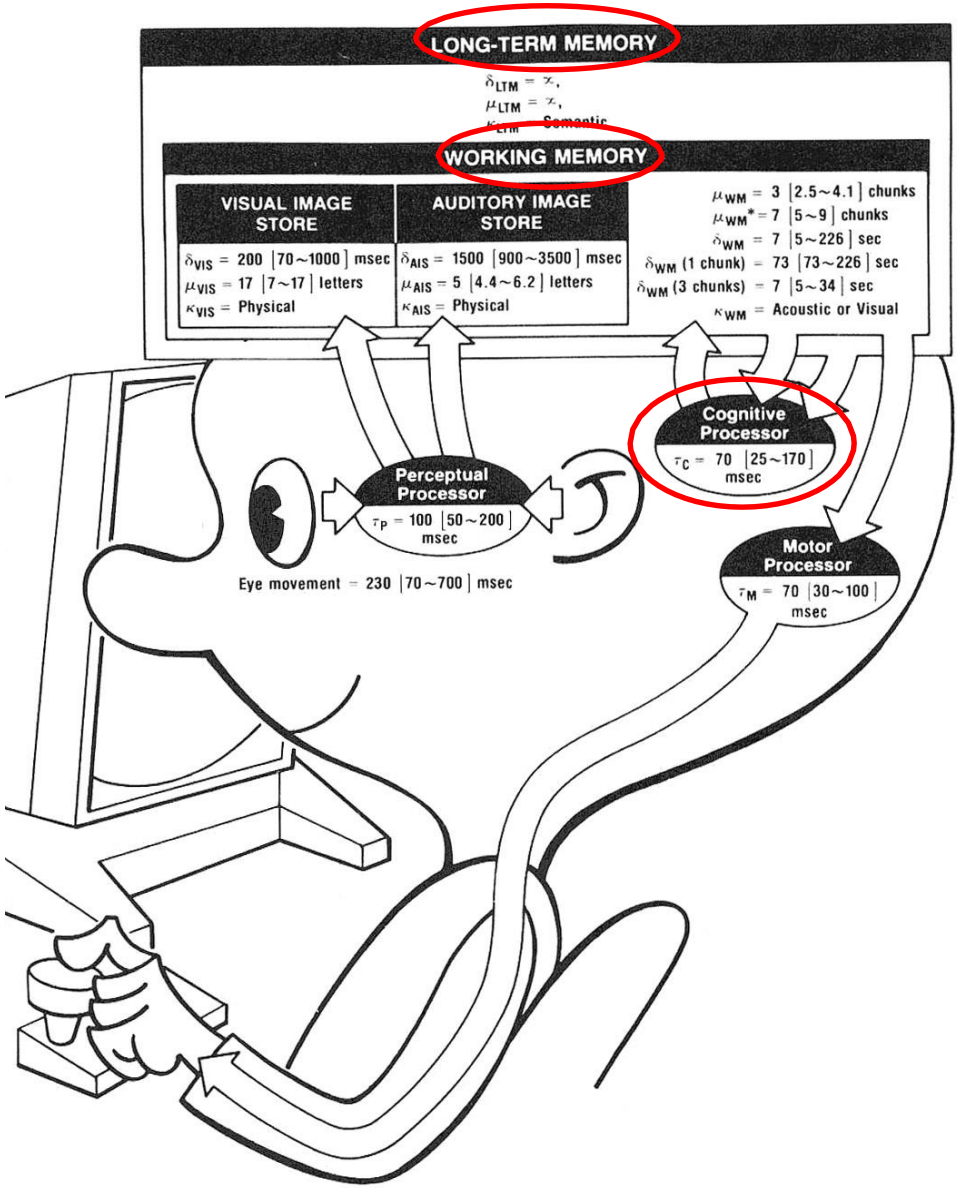
$$\text{Frame rate} > 1/\tau_p = 1 / (100 \text{ ms / frame})$$
$$= 10 \text{ frames / sec}$$

Motor Processor



τ_M=70 [30~100] ms

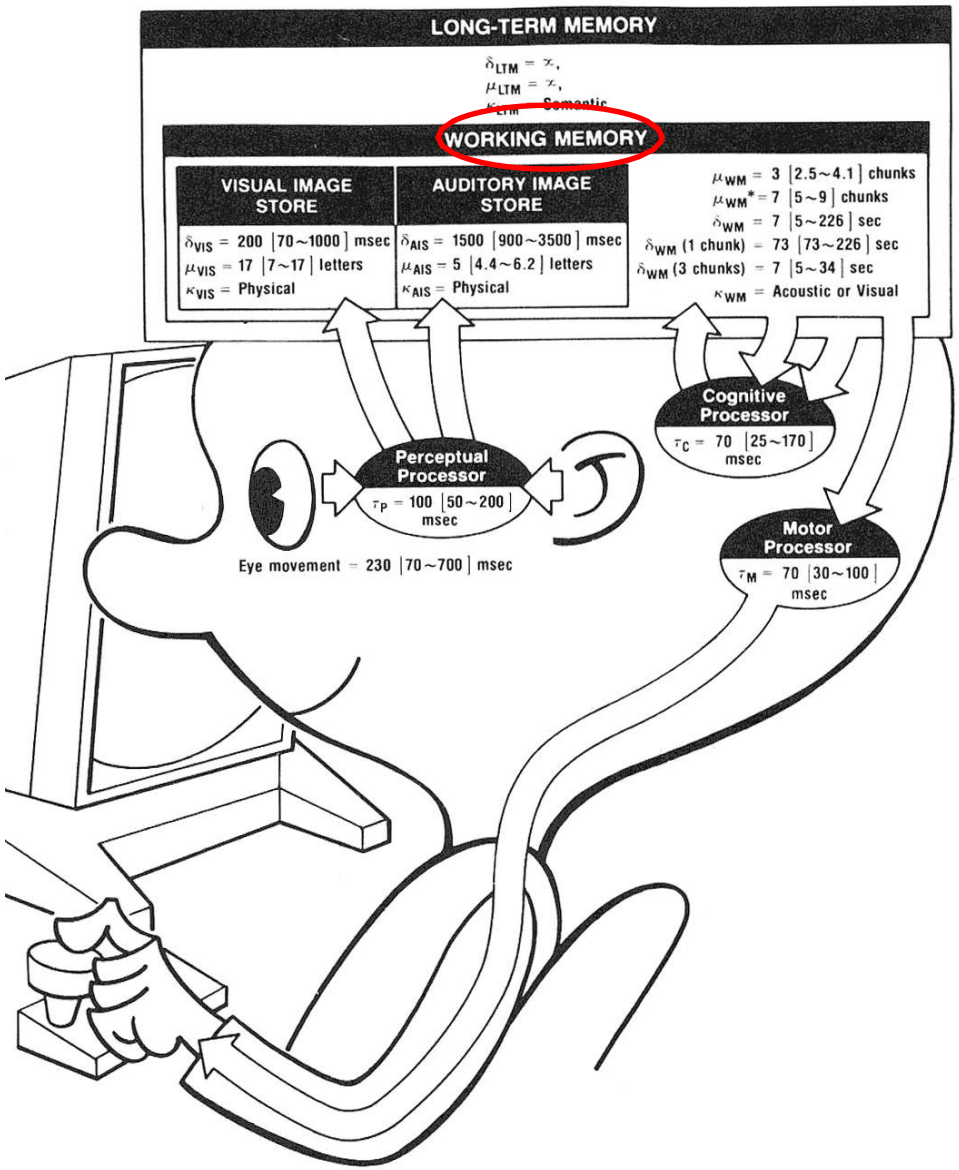
Cognitive System



- Working Memory:
To hold Information under current consideration

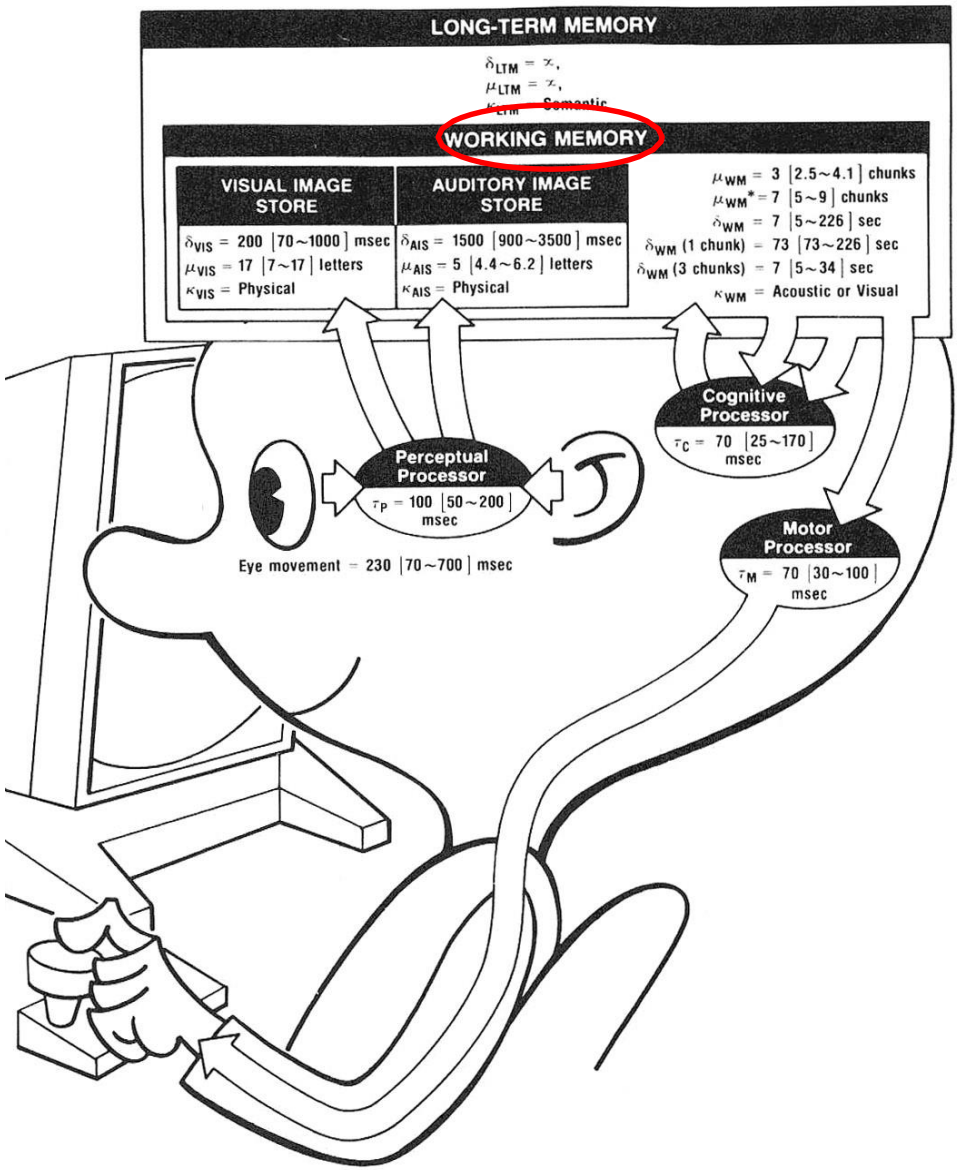
- Long-Term Memory:
To store knowledge for future use

Cognitive System



- Working Memory:
To hold Information under current consideration

Cognitive System



- Working Memory:
To hold Information under current consideration

Consists of *chunks*

Which sequence is easy to repeat?

B C S I B S H A U

Which sequence is easy to repeat?

Which sequence is easy to repeat?

C B S S B I H U A

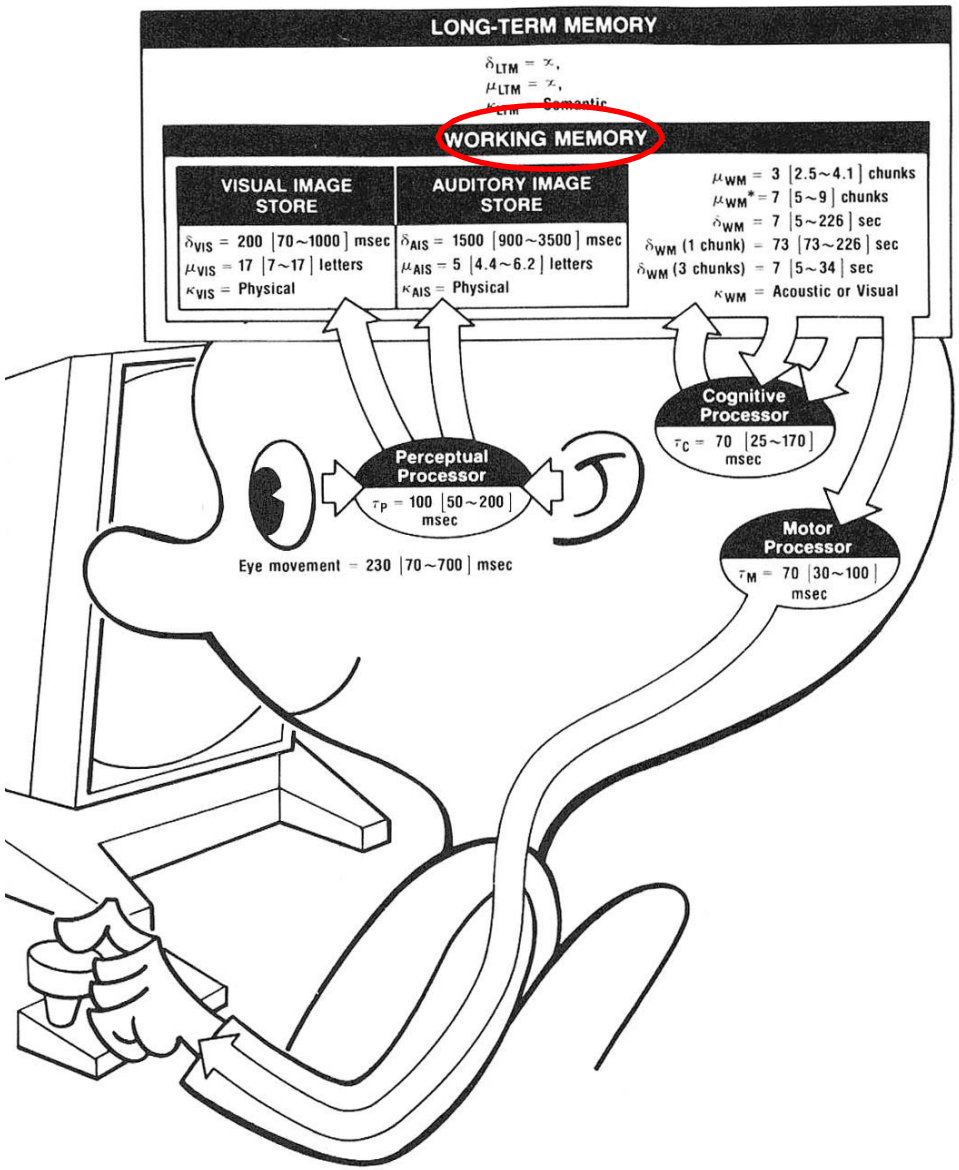
Which sequence is easy to repeat?

Which sequence is easy to repeat?

C B S S B I A H U

C B S	S B I	A H U

Cognitive System



- Working Memory:

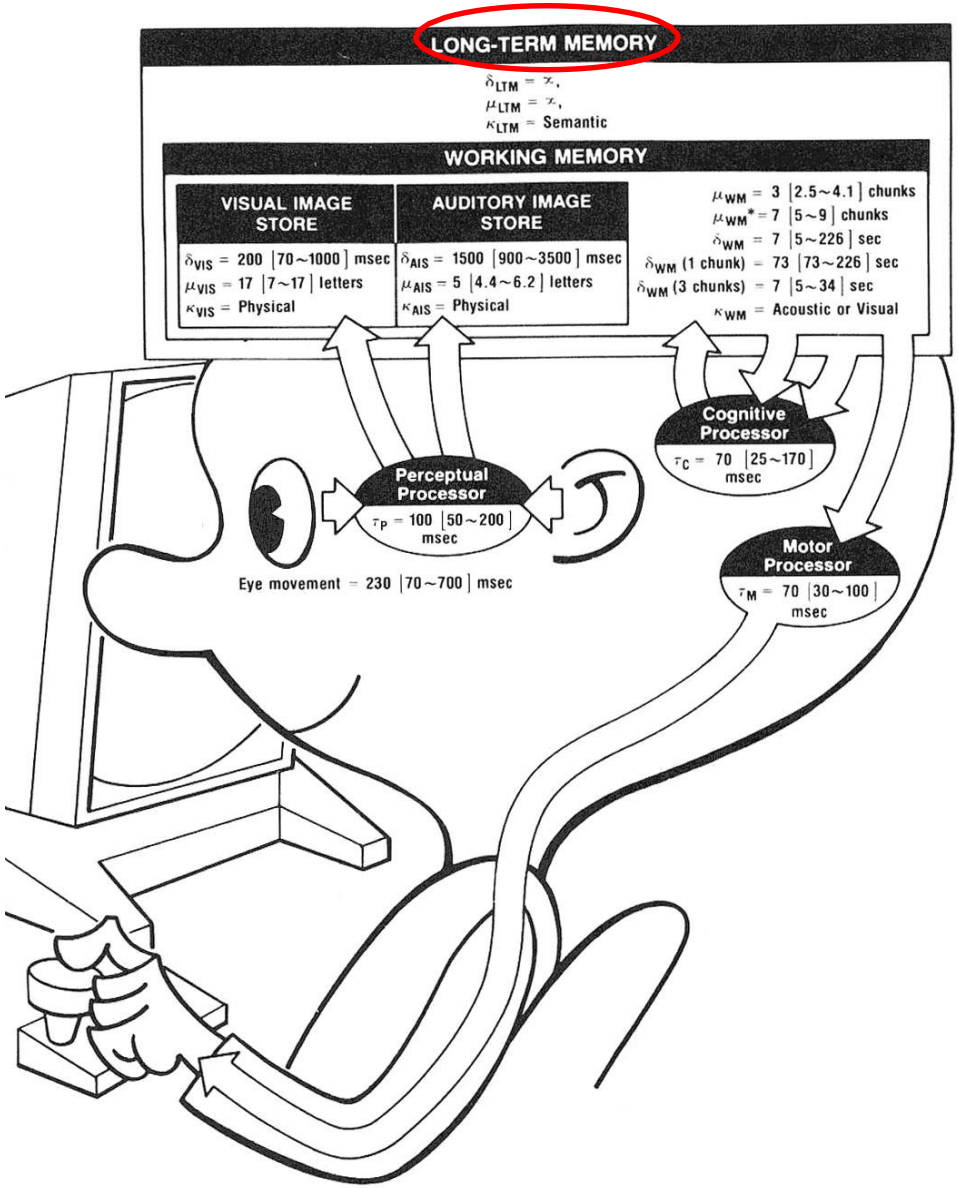
To hold Information under current consideration

δ_{WM}=7[5~226]sec

δ_{WM}(1 chunk) = 73[73~226] sec
δ_{WM}(3 chunk) = 7[5~34] sec

μ_{WM}=7 [5~9] letters

Cognitive System

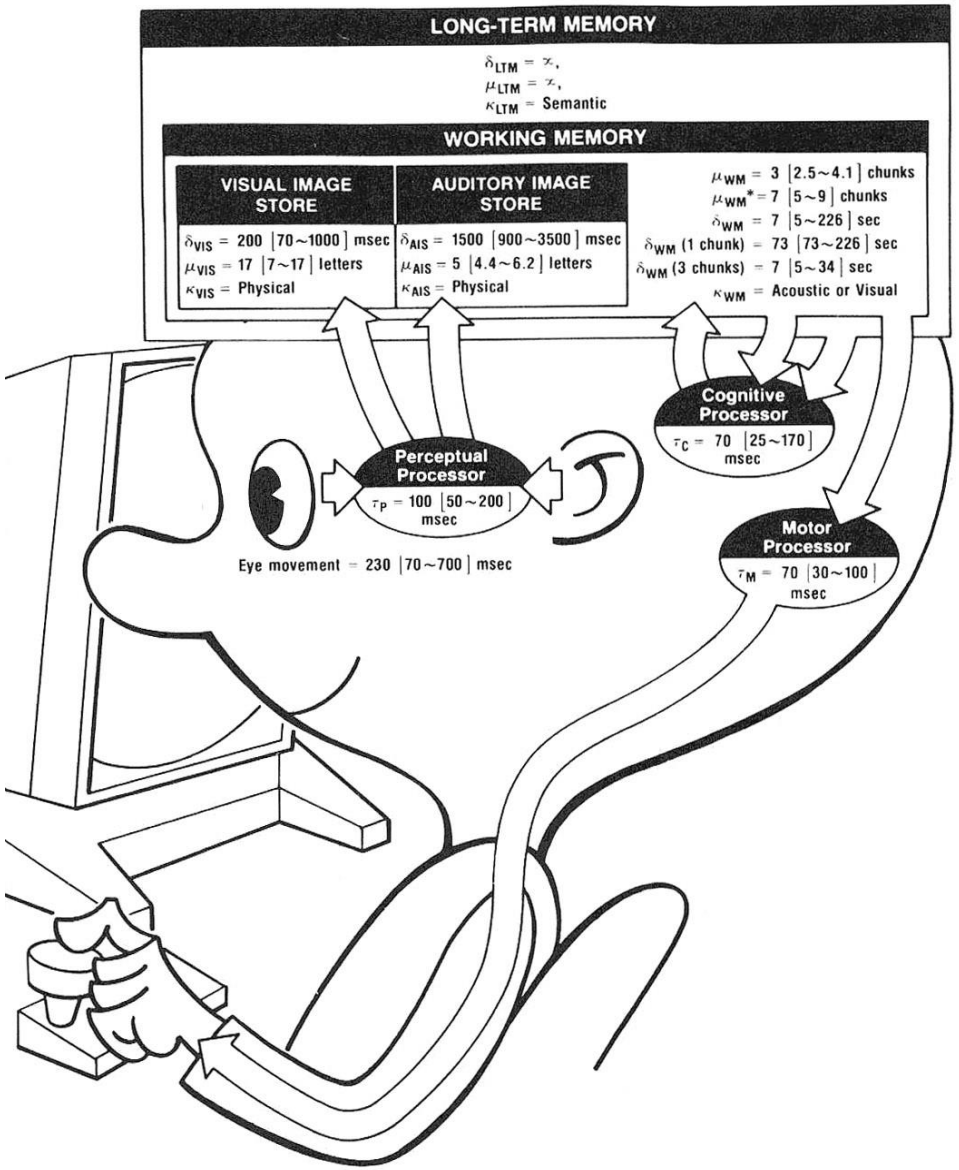


- Long-Term Memory:
  To store knowledge for future use

- How to improve the probability of being retrieved?

Associate it with items in LTM already

Summary



3 subsystems

- Perceptual
- Cognitive
- Motor

- Each subsystem has its own memories and processors.

- Memory
  μ, storage capacity in items
  δ, decay time of an item
  κ, main code type (physical, acoustic, visual, semantic)

- Processor
  τ, cycle time

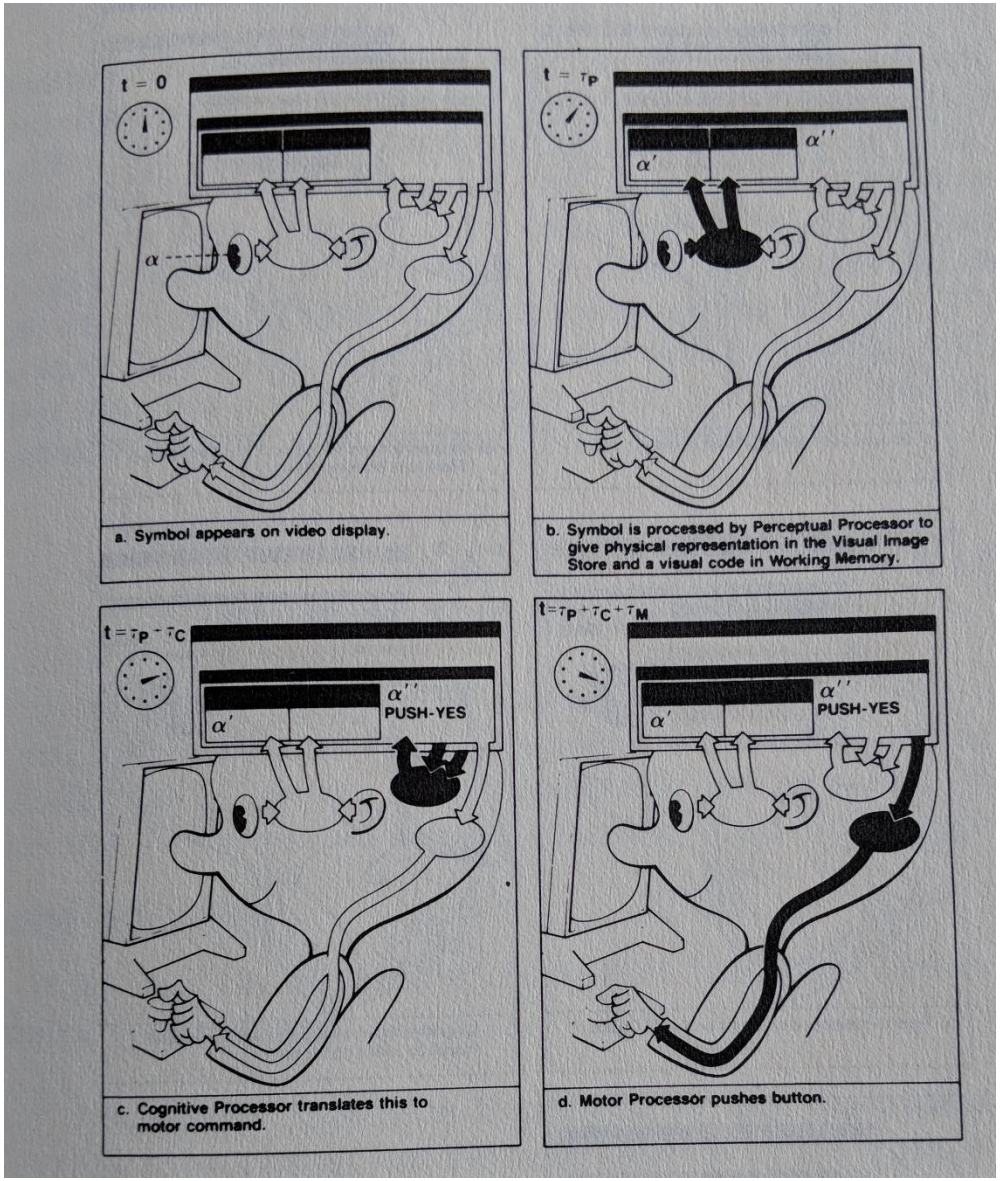
- Three subsystems can work in parallel.

Simple Decisions

Questions

A user sits before a computer display terminal. Whenever any symbol appears, he is to press the space bar. What is the time between signal and response?

Questions



Reaction Time =

$\tau_p + \tau_C + \tau_M =$

$100 + 70 + 70 = 240$ ms

Physical Matches

The user is presented with two symbols, one at a time. If the second symbol is identical to the first, he is to push the key labeled YES, otherwise he is to push NO. What is the time between signal and response for the YES case?

Physical Matches

The user is presented with two symbols, one at a time. If the second symbol is identical to the first, he is to push the key labeled YES, otherwise he is to push NO. What is the time between signal and response for the YES case?

Reaction Time = $\tau_p + 2\tau_C + \tau_M$

$= 100\ [50 \sim 200] + 2 \times (70\ [25 \sim 170]) + 70\ [30 \sim 100]$

$= 310\ [130 \sim 640]$ ms

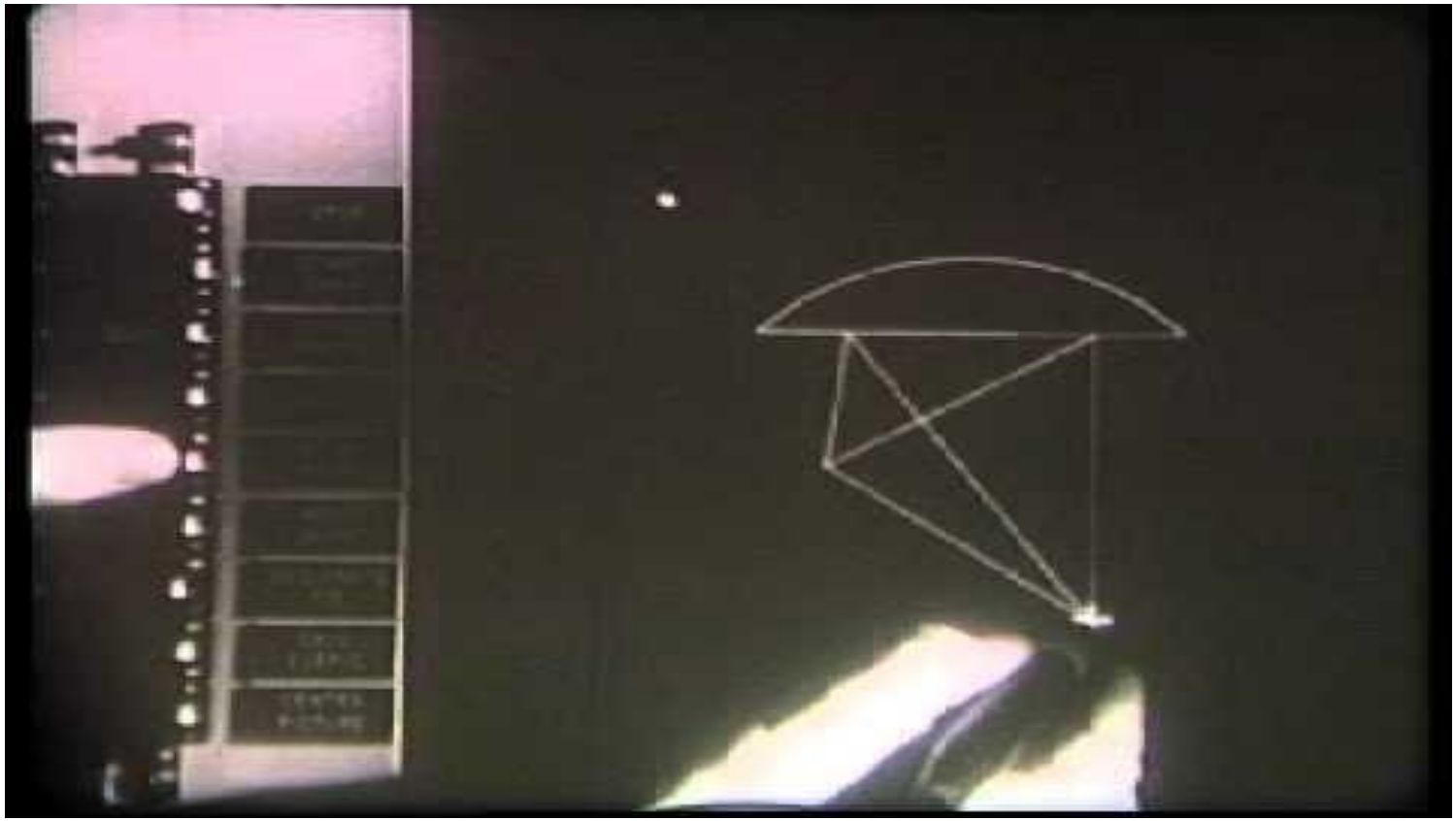
Seminal Ideas

Demo Videos

1968 Englebart's **NLS** Demo

Demo Videos

1963 Sutherland's **SketchPad**



Moving on from the user’s mental model to movement in interactions

Today's Question

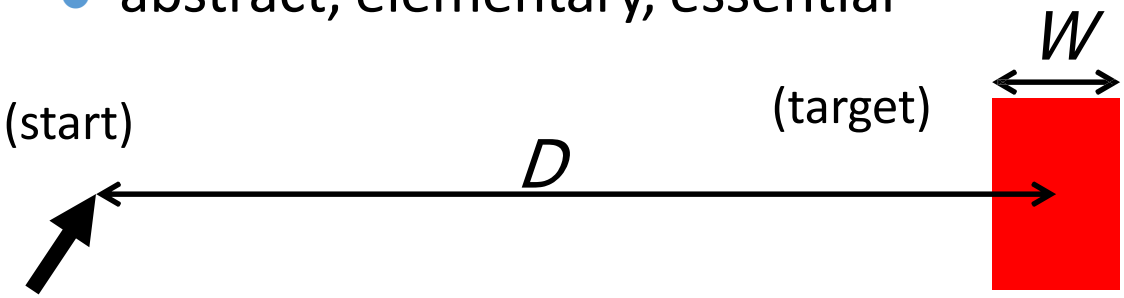
- *You have a palette of tools in a* graphics application that consists of a matrix of 20x16-pixel icons laid out as a 2x8 array that lies along the left- hand edge of the screen. Without moving the array from the left-hand side of the screen or changing the size of the icons, what steps can you take to decrease the time necessary to access the average tool?

Motor Control Models

Pointing Tasks

- Real task: interacting with GUI's
  - Pointing is fundamental

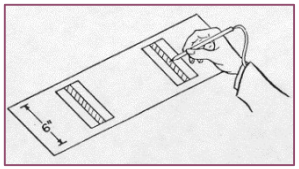
- Experimental task: target acquisition
  - abstract, elementary, essential

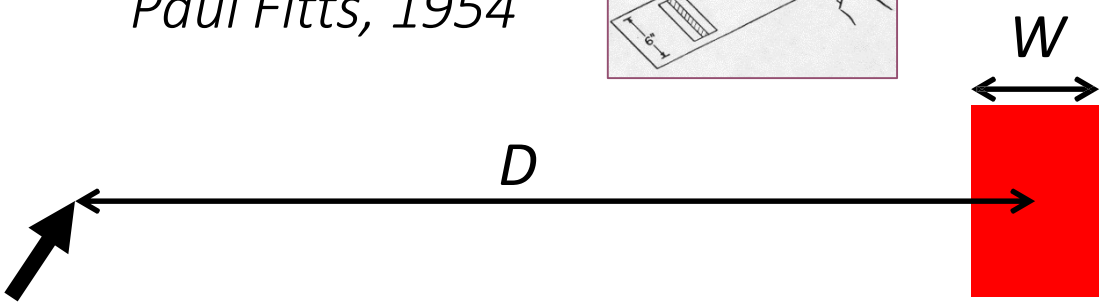


D = distance
W = target width

Fitts' Law

Paul Fitts, 1954

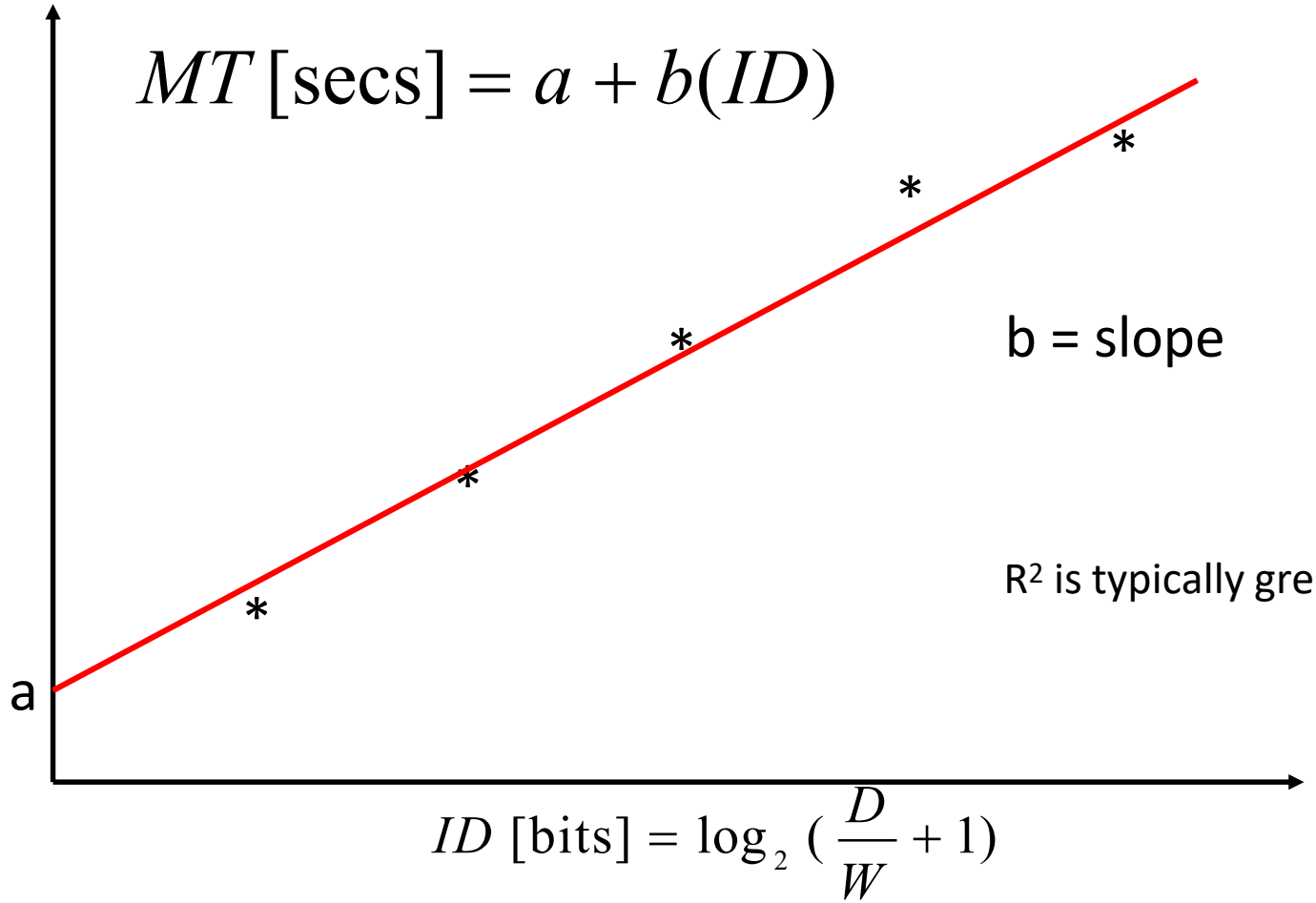




$$\underbrace{MT}_{\textbf{Movement Time}} = a + b\underbrace{\log_2\left(\frac{D}{W}+1\right)}_{\textbf{Index of Difficulty}\ (ID\ [bits])}$$

The original Fitts' law proposed by Paul Fitts in 1954:

$$MT = a + b\ \log_2\left(\frac{2D}{W}\right)$$



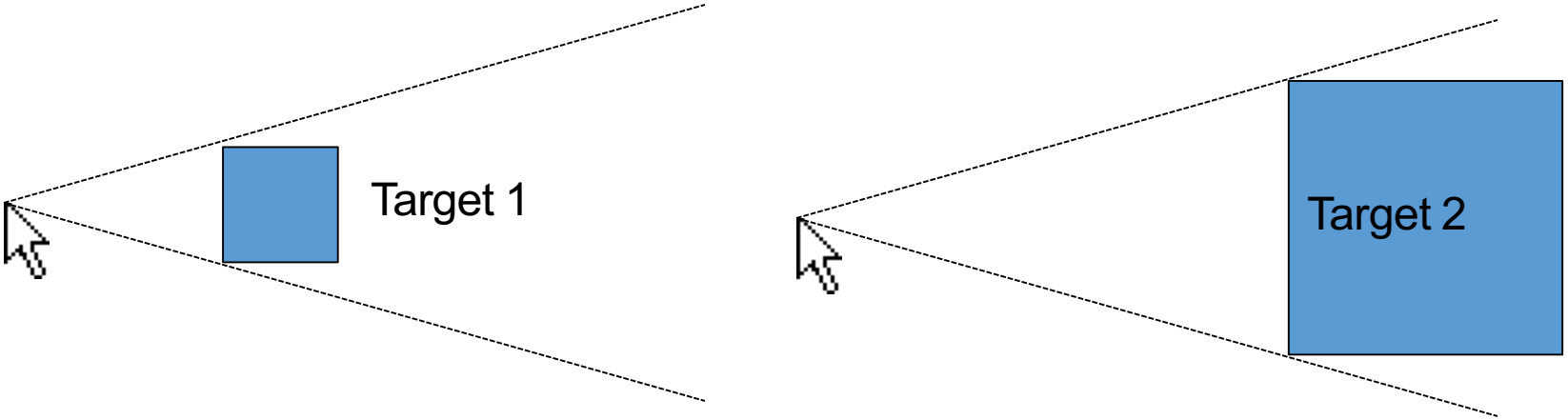
R^2 is typically greater than 0.90

$$MT = a + b\log_2\left(\frac{D}{W} + 1\right)$$

Only Relative Precision (D / W) Matters

Fitts' Law

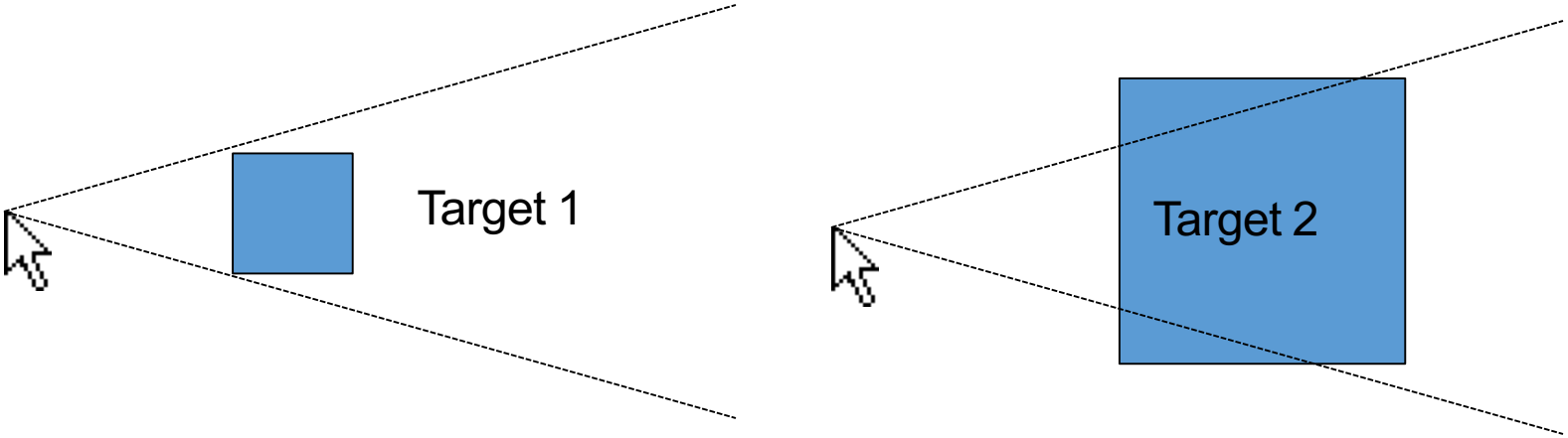
$$\text{Time} = a + b \log_2(D/W+1)$$



Same ID → Same Difficulty

Fitts' Law

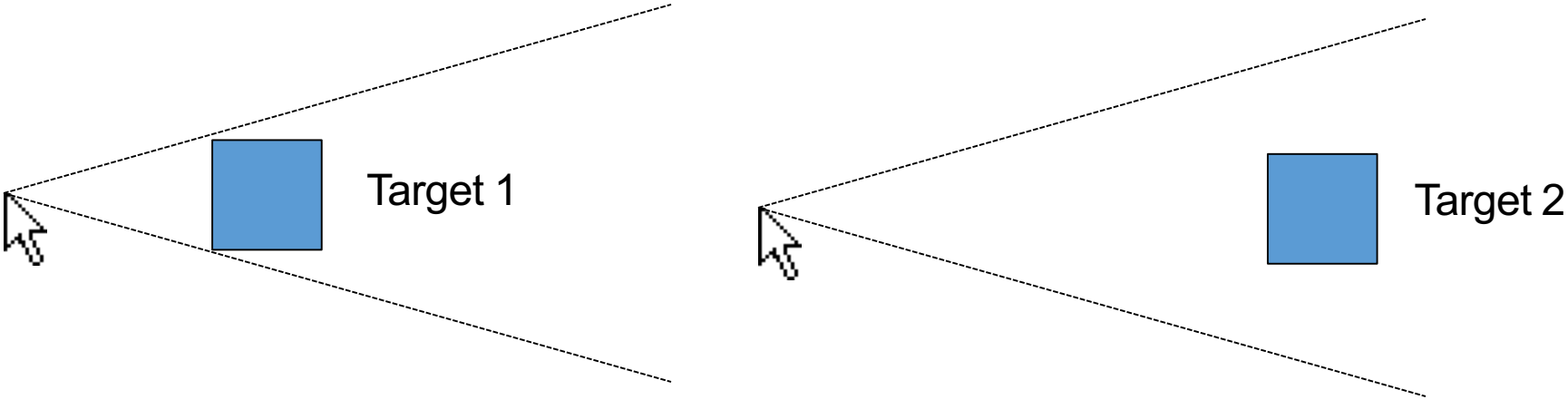
$$\text{Time} = a + b \log_2(D/W+1)$$



Smaller ID → Easier

Fitts' Law

$$\text{Time} = a + b \log_2(D/W+1)$$



Larger ID → Harder

Bandwidth (IP) of Human Muscle Groups

Index of Performance

$IP = 1/b$

Which is faster?

Mouse

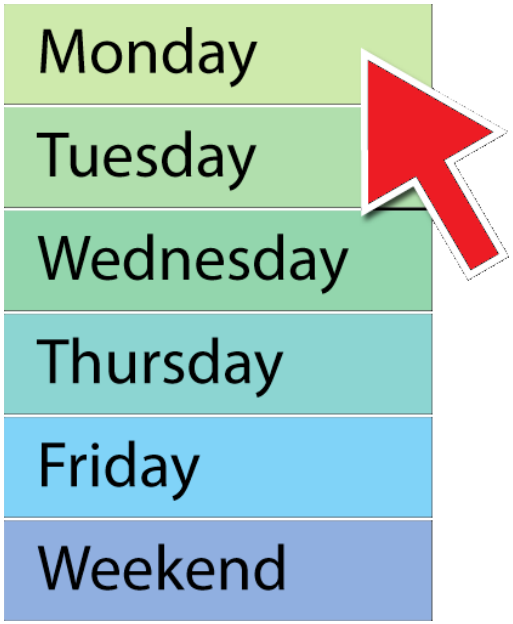
Joystick

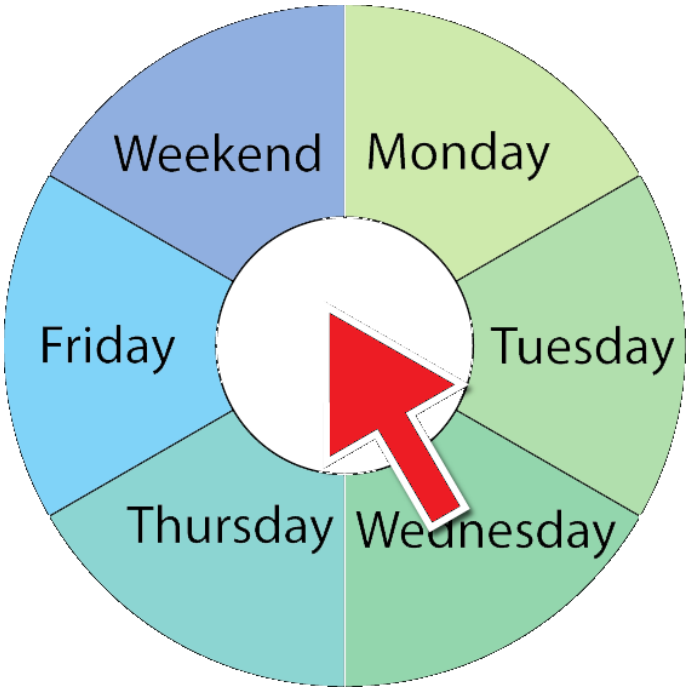
50 years of data

Device	Study	*IP* (bits/s)
Hand	Fitts (1954)	10.6
Mouse	Card, English, & Burr (1978)	10.4
Joystick	Card, English, & Burr (1978)	5.0
Trackball	Epps (1986)	2.9
Touchpad	Epps (1986)	1.6
Eyetracker	Ware & Mikaelian (1987)	13.7

Reference:
MacKenzie, I. Fitts' Law as a research and design tool in human computer interaction. *Human Computer Interaction, 1992, Vol. 7, pp. 91-139*

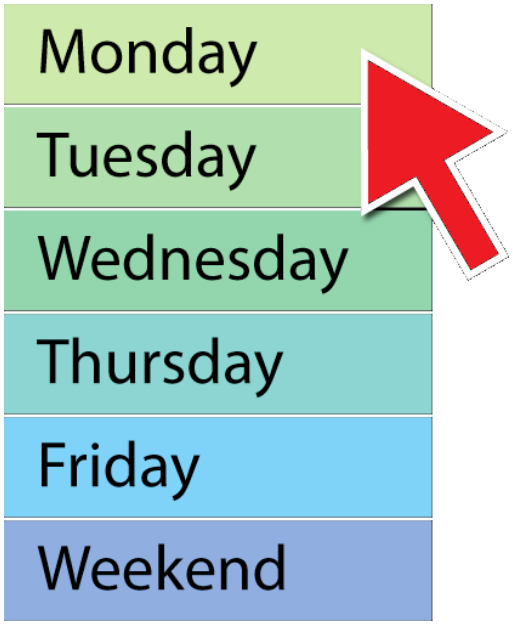
Fitts' Law Example

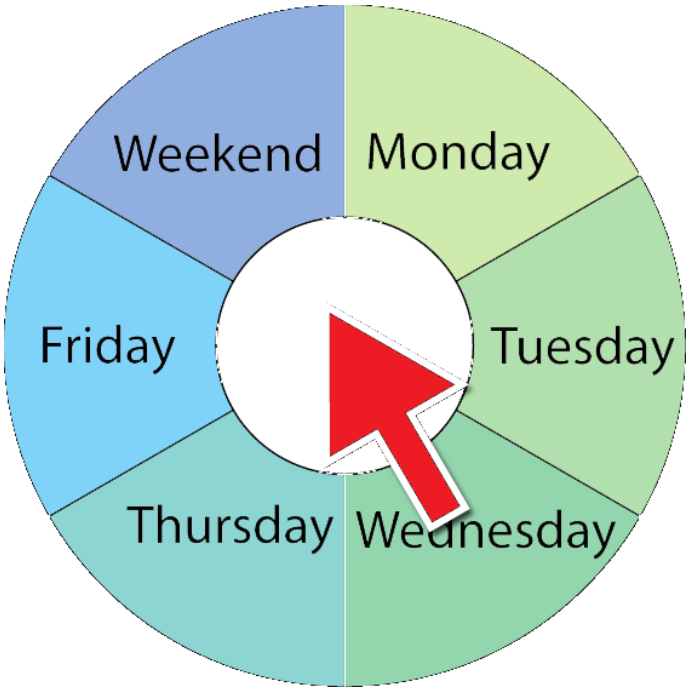




Which will be faster on average?

Fitts' Law Example





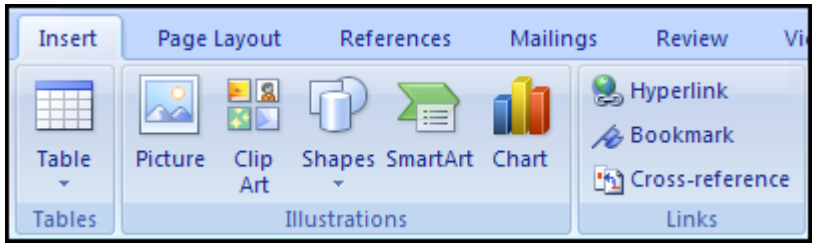
Which will be faster on average?

Pie menu (bigger targets & less distance)

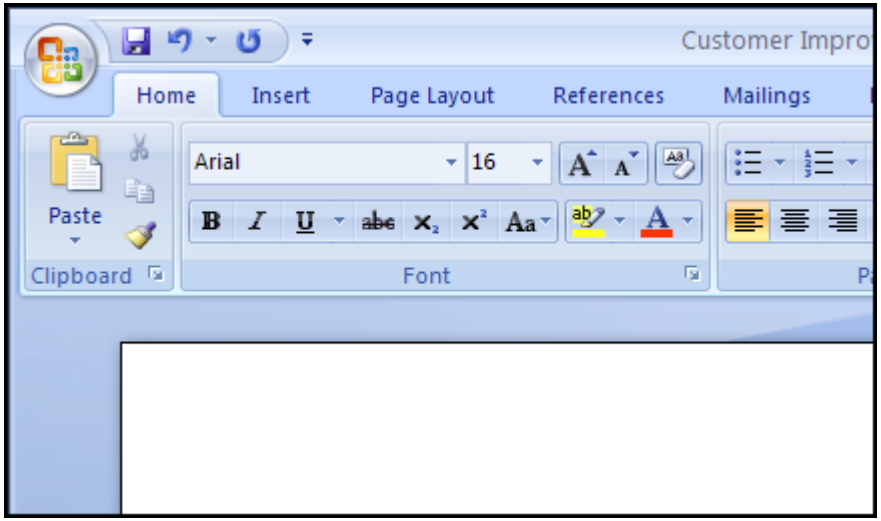
Fitts' Law in Windows and Mac OS

Windows 95: Missed by a pixel

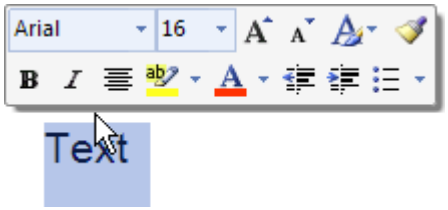
Fitts' Law in Office 2007



Larger, labeled controls can be clicked more quickly



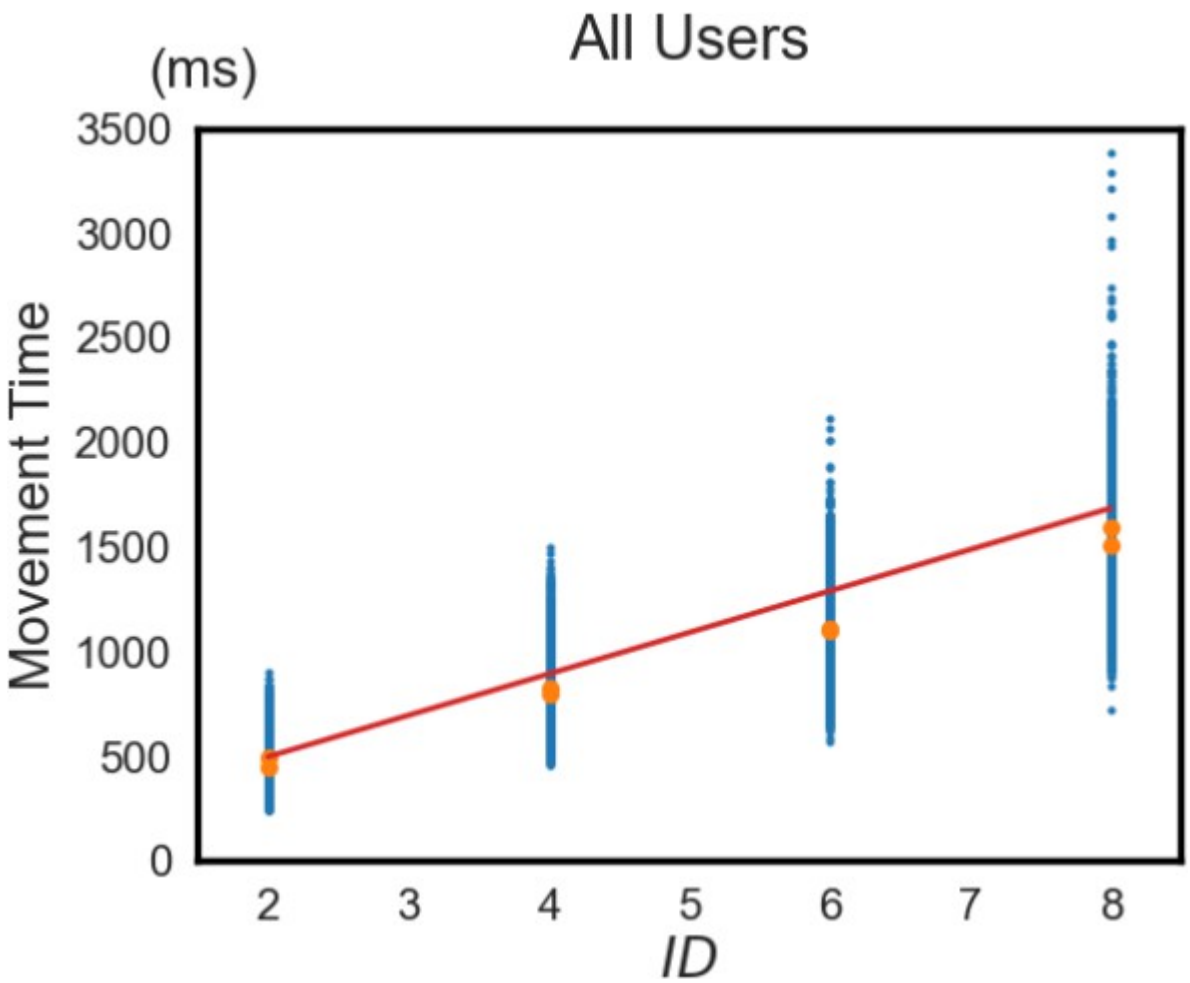
Magic Corner: Office Button in the upper-left corner



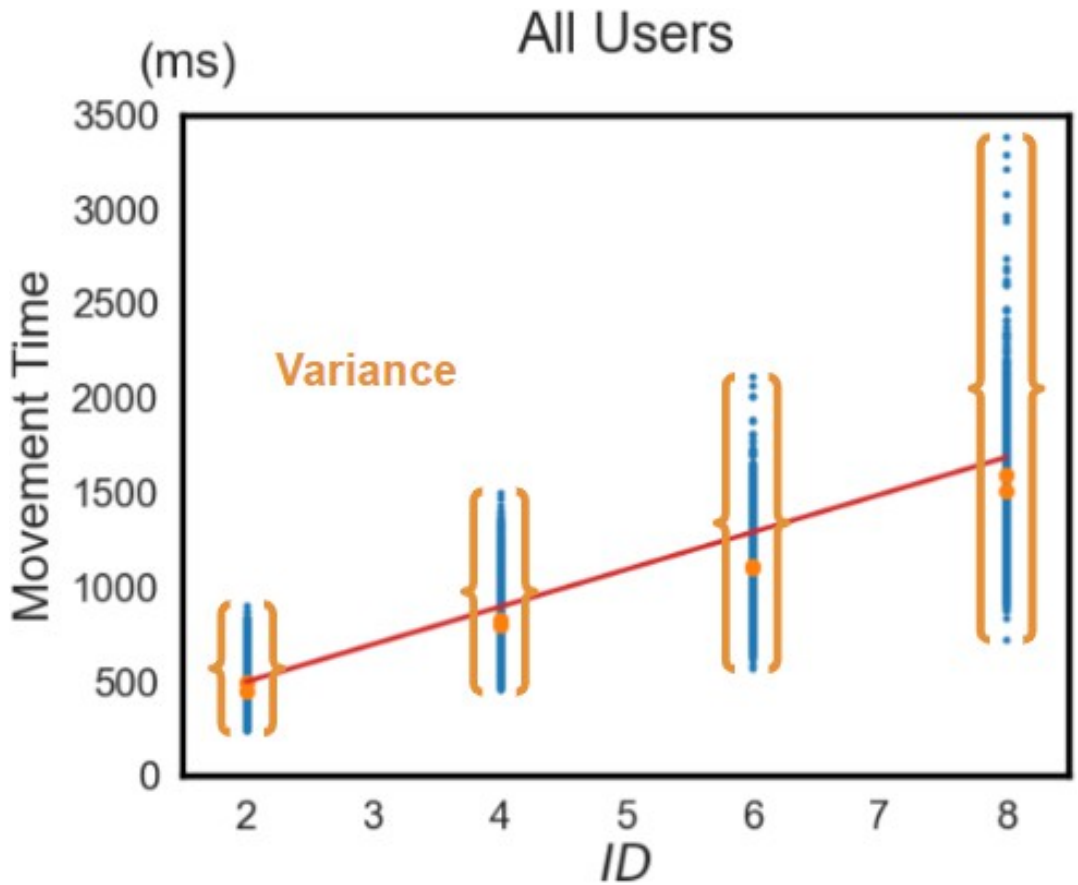
Mini Toolbar: Close to the cursor

Source: Jensen Harris, An Office User Interface Blog : Giving You Fitts. Microsoft, 2007.

Distribution of MT



$$ID = \log_2\left(\frac{D}{W} + 1\right)$$



$$V : \text{variance}, V = c + d \cdot ID^2$$

$$ID = \log_2\left(\frac{D}{W} + 1\right)$$

2D Fitts' Law

- Fitts' Law is for 1D. How about 2D?
  - Extending Fitts' law to two-dimensional tasks (CHI 92', Mackenzie and Buxton)
    - 5 simple model proposed, 2 of them outperform according to the experiment

Extending Fitts' law to Two-Dimensional Tasks

- Original Fitts' Law

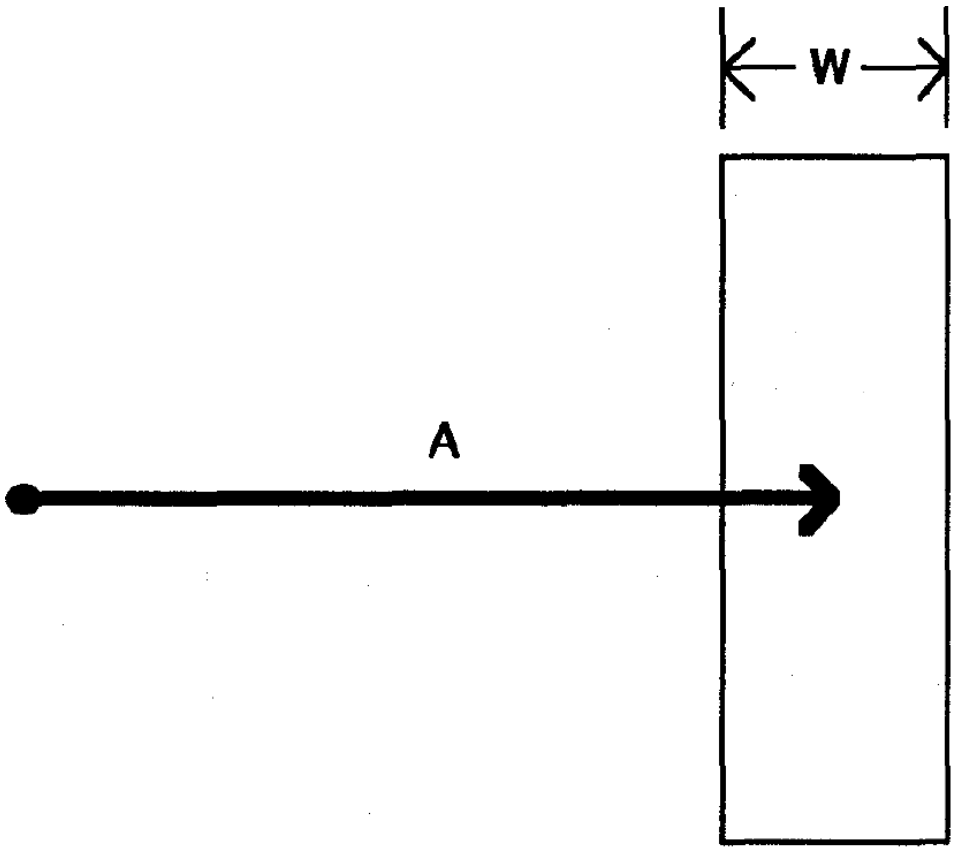
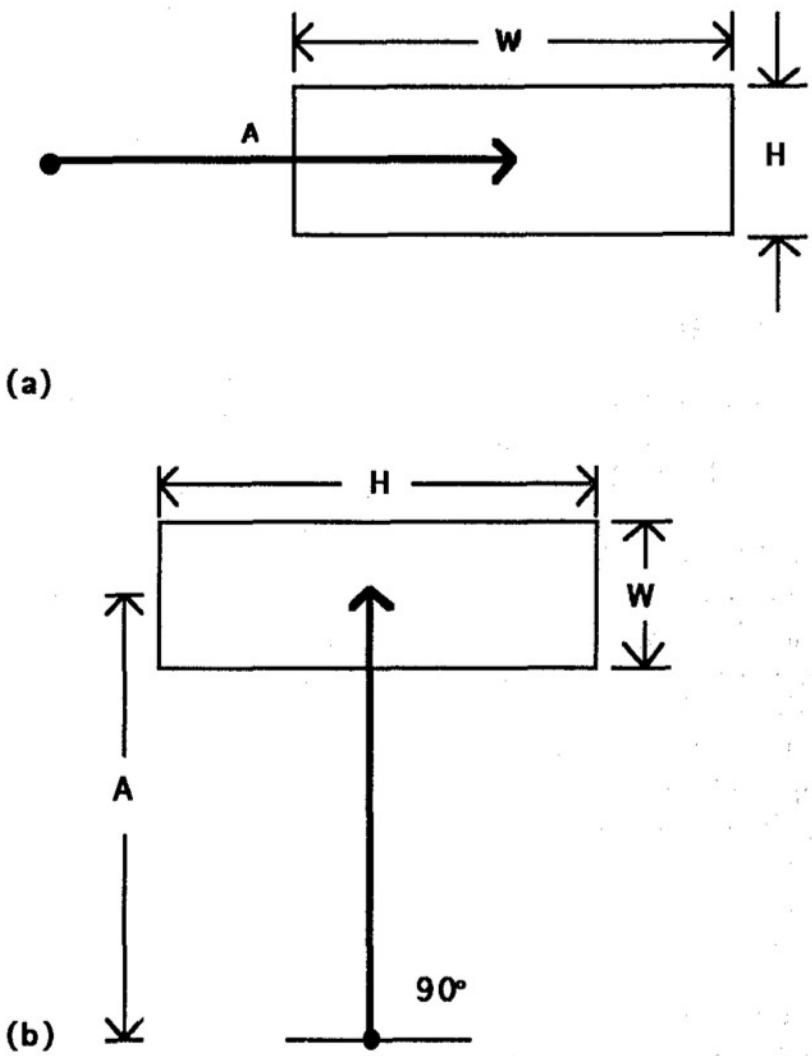


Figure 1. Fitts' law paradigm. The law is inherently one-dimensional since target amplitude (A) and width (W) are measured along the same axis.

Extending Fitts' law to Two-Dimensional Tasks

- Which one should we take into account? Width? Or Height?



Extending Fitts' law to Two-Dimensional Tasks

- How about the cases which are not horizontal or vertical?

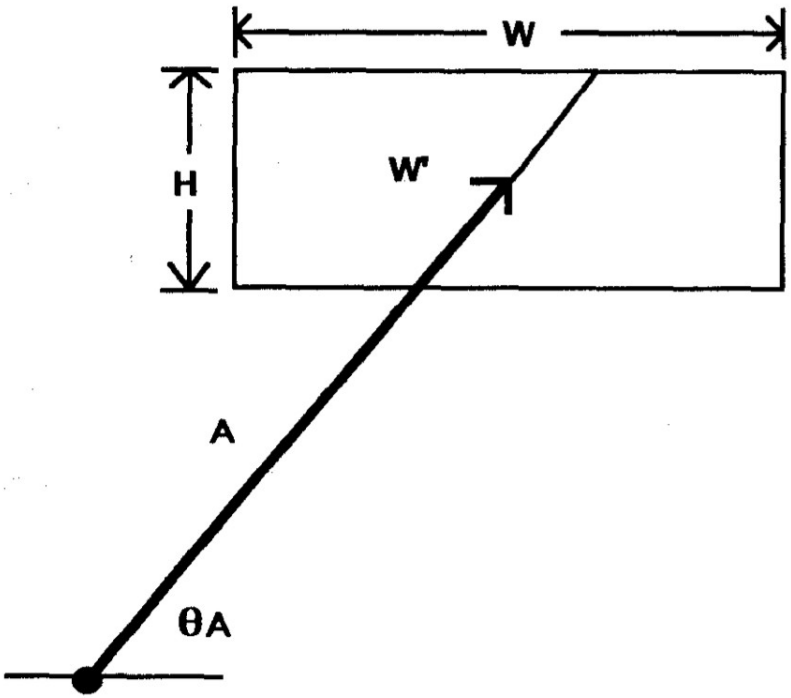


Figure 3. What is target width? Possibilities include W' (the width of the target along an approach vector) or the smaller of W or H.

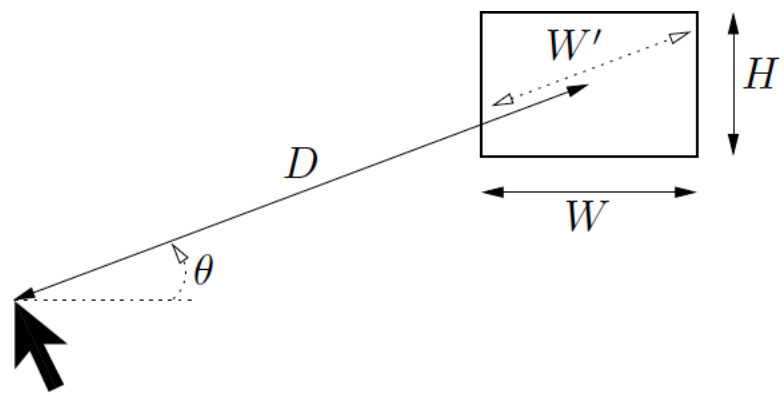
Extending Fitts' law to Two-Dimensional Tasks

- 5 model proposed
  - "STATUS QUO" model
  - "SMALLER OF" model
  - Apparent width W' model
  - Substitute W with W*H
  - Substitute W with W+H

$$MT = a + b\log_2\left(\frac{D}{W} + 1\right)$$

$$ID_{\min(W,H)} = \log_2\left(\frac{D}{\min(W,H)} + 1\right)$$

$$ID_{W'} = \log_2\left(\frac{D}{W'} + 1\right)$$



Extending Fitts' law to Two-Dimensional Tasks

- Experiment Result

Empirically, this is the best

Model for Target Width	*ID* Range (bits) Low	*ID* Range (bits) High	r^a	SE^b (ms)	Regression Coefficients: Intercept, a (ms)	Regression Coefficients: Slope, b (ms/bit)	Regression Coefficients: *IP* (bits/s)
SMALLER-OF	1.58	5.04	.9501	64	230	166	6.0
W	1.00	5.04	.9333	74	337	160	6.3
$W+H$	0.74	3.54	.8755	99	402	218	4.6
$W \times H$	0.32	4.09	.8446	110	481	173	5.8
STATUS QUO	1.00	5.04	.8097	121	409	135	7.4

[a] $n = 78, p < .001$

[b] standard error of estimate

Theoretically, this is similar to Fitts' Law

Figure 6. Correlations and regression coefficients for five models for target width.

Today's Questions

- *You have a palette of tools in a* graphics application that consists of a matrix of 20x16-pixel icons laid out as a 2x8 array that lies along the left- hand edge of the screen. Without moving the array from the left-hand side of the screen or changing the size of the icons, what steps can you take to decrease the time necessary to access the average tool?

1. Change the array to 1X16, so all the tools lie along the edge of the screen.

2. Ensure that the user can click on the very first row of pixels along the edge of the screen to select a tool. There should be no buffer zone.

Questions?